In [1]:
```
# Initialize Otter
import otter
grader = otter.Notebook("hw5.ipynb")
```

CPSC 330 - Applied Machine Learning

Homework 5: Putting it all together

Associated lectures: All material till lecture 13

Instructions

rubric={points}

You will earn points for following these instructions and successfully submitting your work on Gradescope.

Before you start

- Please **read carefully Use of Generative AI policy** before starting the homework assignment.
- Review the CPSC 330 homework instructions for detailed guidance on completing and submitting assignments.

Group work instructions

You may work with a partner on this homework and submit your assignment as a group. Below are some instructions on working as a group.

- The maximum group size is 2.
- Use group work as an opportunity to collaborate and learn new things from each other.
- Be respectful to each other and make sure you understand all the concepts in the assignment well.
- It's your responsibility to make sure that the assignment is submitted by one of the group members before the deadline.
- You can find the instructions on how to do group submission on Gradescope here.

- If you would like to use late tokens for the homework, all group members must have the necessary late tokens available. Please note that the late tokens will be counted for all members of the group.

Before submitting

- **Run all cells** (▶▶ button) to ensure the notebook executes cleanly from top to bottom.
  - Execution counts must start at **1** and be sequential.
  - Notebooks with missing outputs or errors may lose marks.
- **Do not upload or push data files** used in this lab to GitHub or Gradescope. (A `.gitignore` is provided to prevent this.)

Submitting on Gradescope

- Upload **only** your `.ipynb` file (with outputs shown) and any required output files. Do **not** submit extra files.
- If needed, refer to the Gradescope Student Guide.
- If your notebook is too large to render, also upload a **Web PDF** or **HTML** version.
  - You can create one using **File** → **Save and Export Notebook As**.
  - If you get an error when creating a PDF, try running the following commands in your lab directory:

    ```
    conda install -c conda-forge nbconvert-playwright
    jupyter nbconvert --to webpdf lab1.ipynb
    ```
  - Ensure all outputs are visible in your PDF or HTML file; TAs cannot grade your work if outputs are missing.

Note: Unlike previous assignments, this one is open-ended and project-style. Treat it as an opportunity to explore, experiment, and learn.

Imports

Imports

Points: 0

In [2]:
```
import pandas as pd
```

Introduction

In this homework you will be working on an open-ended mini-project, where you will put all the different things you have learned so far together to solve an interesting problem.

A few notes and tips when you work on this mini-project:

Tips

1. This mini-project is open-ended, and while working on it, there might be some situations where you'll have to use your own judgment and make your own decisions (as you would be doing when you work as a data scientist). Make sure you explain your decisions whenever necessary.
2. **Do not include everything you ever tried in your submission** -- it's fine just to have your final code. That said, your code should be reproducible and well-documented. For example, if you chose your hyperparameters based on some hyperparameter optimization experiment, you should leave in the code for that experiment so that someone else could re-run it and obtain the same hyperparameters, rather than mysteriously just setting the hyperparameters to some (carefully chosen) values in your code.
3. If you realize that you are repeating a lot of code try to organize it in functions. Clear presentation of your code, experiments, and results is the key to be successful in this lab. You may use code from lecture notes or previous lab solutions with appropriate attributions.

Assessment

We plan to grade fairly and leniently. We don't have some secret target score that you need to achieve to get a good grade. **You'll be assessed on demonstration of mastery of course topics, clear presentation, and the quality of your analysis and results.** For example, if you just have a bunch of code and no text or figures, that's not good. If you do a bunch of sane things and get a lower accuracy than your friend, don't sweat it.

A final note

Finally, this style of this "project" question is different from other assignments. It'll be up to you to decide when you're "done" -- in fact, this is one of the hardest parts of real projects. But please don't spend WAY too much time on this... perhaps "a few hours" (10-14 hours???) is a good guideline for this project . Of course if you're having fun you're welcome to spend as much time as you want! But, if so, try not to do it out of perfectionism or getting the best possible grade. Do it because you're learning and enjoying it. Students from the past cohorts have found such kind of labs useful and fun and I hope you enjoy it as well.

1. Pick your problem and explain the prediction problem

rubric={points:3}
In this mini project, you have the option to choose on which dataset you will be working on. The tasks you will need to carry on will be similar, independently of your choice.

Option 1

You can choose to work on a classification problem of predicting whether a credit card client will default or not. For this problem, you will use Default of Credit Card Clients Dataset. In this data set, there are 30,000 examples and 24 features, and the goal is to estimate whether a person will default (fail to pay) their credit card bills; this column is labeled "default.payment.next.month" in the data. The rest of the columns can be used as features. You may take some ideas and compare your results with the associated research paper, which is available through the UBC library.

Option 2

You can choose to work on a regression problem using a dataset of New York City Airbnb listings from 2019. As usual, you'll need to start by downloading the dataset, then you will try to predict `reviews_per_month`, as a proxy for the popularity of the listing. Airbnb could use this sort of model to predict how popular future listings might be before they are posted, perhaps to help guide hosts create more appealing listings. In reality they might instead use something like vacancy rate or average rating as their target, but we do not have that available here.

> Note there is an updated version of this dataset with more features available here. The features were are using in `listings.csv.gz` for the New York city datasets. You will also see some other files like `reviews.csv.gz`. For your own interest you may want to explore the expanded dataset and try your analysis there. However, please submit your results on the dataset obtained from Kaggle.

Your tasks:

1. Review the available datasets and choose the one you find most interesting. It may help to read through the dataset documentation on Kaggle before deciding.
2. Once you've selected a dataset, take time to understand the problem it represents and the meaning of each feature. Use the Kaggle documentation to guide you.
3. Download the dataset and load it into a pandas DataFrame.
4. Write a few sentences summarizing your initial thoughts about the problem and the dataset.

Solution_1

Points: 3

Type your answer here, replacing this text.

In [3]:
```
# Load Option 1 dataset:
df = pd.read_csv("UCI_Credit_Card.csv")
df.head()
```

Out[3]:

	ID	LIMIT_BAL	SEX	EDUCATION	MARRIAGE	AGE	PAY_0	PAY_2	PAY_3	PAY_4	...	BILL_AMT4	BILL_AMT5	BILL_AMT6	PAY_A
0	1	20000.0	2	2	1	24	2	2	-1	-1	...	0.0	0.0	0.0	
1	2	120000.0	2	2	2	26	-1	2	0	0	...	3272.0	3455.0	3261.0	
2	3	90000.0	2	2	2	34	0	0	0	0	...	14331.0	14948.0	15549.0	15
3	4	50000.0	2	2	1	37	0	0	0	0	...	28314.0	28959.0	29547.0	20
4	5	50000.0	1	2	1	57	-1	0	-1	0	...	20940.0	19146.0	19131.0	20

5 rows × 25 columns

In [4]:
```
#Glance at the Data:
print("Shape:", df.shape)
print("\nColumns:")
print(df.columns.tolist())
```

```
Shape: (30000, 25)

Columns:
['ID', 'LIMIT_BAL', 'SEX', 'EDUCATION', 'MARRIAGE', 'AGE', 'PAY_0', 'PAY_2', 'PAY_3', 'PAY_4', 'PAY_5', 'PAY_6', 'BILL_
AMT1', 'BILL_AMT2', 'BILL_AMT3', 'BILL_AMT4', 'BILL_AMT5', 'BILL_AMT6', 'PAY_AMT1', 'PAY_AMT2', 'PAY_AMT3', 'PAY_AMT4',
'PAY_AMT5', 'PAY_AMT6', 'default.payment.next.month']
```

In [5]:
```
#List out the features and TARGET:
target_col = "default.payment.next.month"
X = df.drop(columns=[target_col])
y = df[target_col]

print("Target column:", target_col)
print("Number of features:", X.shape[1])
print("Target distribution:")
print(y.value_counts())
```

```python
print("\nTarget proportion:")
print(y.value_counts(normalize=True).round(4))
```

```
Target column: default.payment.next.month
Number of features: 24
Target distribution:
default.payment.next.month
0    23364
1     6636
Name: count, dtype: int64

Target proportion:
default.payment.next.month
0    0.7788
1    0.2212
Name: proportion, dtype: float64
```

Comments:

I chose Option 1, the Default of Credit Card Clients dataset for this mini-project. This is a binary classification problem where the goal is to predict whether a credit card client will default on their payment in the next month. In the dataset, the target variable is `default.payment.next.month`, where a value of 1 indicates default and 0 indicates no default.

This problem is meaningful because predicting default risk can help financial institutions make better decisions about lending, risk management, and customer support (That sounds pretty fun to me). The dataset includes customer demographic information such as age, sex, education, and marital status. Also credit limit information, repayment status history, bill amounts, and payment amounts. These variables can be used as features to predict the target.

I loaded the dataset into a pandas. Since the target column is already provided and the data is relatively clean, this dataset is a good fit for building and comparing supervised machine learning models in a reproducible way.

2. Data splitting

---

rubric={points:2}

Your tasks:

1. Split the data into train (70%) and test (30%) portions with `random_state=123`.

> If your computer cannot handle training on 70% training data, make the test split bigger.

Solution_2

Points: 2

In [6]:
```
from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size = 0.30,
    random_state = 123,
    stratify = y
)

print("X_train shape:", X_train.shape)
print("X_test shape:", X_test.shape)
print("y_train shape:", y_train.shape)
print("y_test shape:", y_test.shape)
```

```
X_train shape: (21000, 24)
X_test shape: (9000, 24)
y_train shape: (21000,)
y_test shape: (9000,)
```

In [7]:
```
print("Original target proportion:")
print(y.value_counts(normalize=True).round(4))
```

```
Original target proportion:
default.payment.next.month
0    0.7788
1    0.2212
Name: proportion, dtype: float64
```

Comments:

I split the data into a training set (70%) and a test set (30%) using train_test_split with random_state=123 as required. I used stratify=y so that the class proportions are approximately preserved in both the training and test sets, since it is a classification problem with an imbalance target.

3. EDA

rubric={points:10}

Your tasks:

1. **Perform exploratory data analysis (EDA)**: Conduct an initial exploration of the training set to better understand its characteristics.
2. **Summarize and visualize the data**: Include at least **two summary statistics** and **two visualizations** that you find informative. For each, write **one sentence** explaining what insight it provides.
3. **Record your observations**: Summarize your **initial observations** about the dataset based on your EDA.
4. **Select evaluation metrics**: Choose one or more **appropriate metrics** for assessing model performance and briefly justify your choice.

Solution_3

Points: 10

Type your answer here, replacing this text.

In [8]:
```
import numpy as np
import matplotlib.pyplot as plt

#Combine training features and target for EDA:
train_df = X_train.copy()
train_df["default.payment.next.month"] = y_train

print("Training set shape:", train_df.shape)
print("\nMissing values (top 10 columns):")
print(train_df.isnull().sum().sort_values(ascending=False).head(10))

print("\nData types:")
print(train_df.dtypes.value_counts())
```

```
Training set shape: (21000, 25)

Missing values (top 10 columns):
ID           0
BILL_AMT2    0
PAY_AMT6     0
PAY_AMT5     0
PAY_AMT4     0
PAY_AMT3     0
PAY_AMT2     0
PAY_AMT1     0
BILL_AMT6    0
BILL_AMT5    0
dtype: int64

Data types:
float64    13
int64      12
Name: count, dtype: int64
```

In [9]:

```python
target_col = "default.payment.next.month"

print("Training target counts:")
print(train_df[target_col].value_counts())

print("\nTraining target proportions:")
print(train_df[target_col].value_counts(normalize=True).round(4))
```

```
Training target counts:
default.payment.next.month
0    16355
1     4645
Name: count, dtype: int64

Training target proportions:
default.payment.next.month
0    0.7788
1    0.2212
Name: proportion, dtype: float64
```

-- The target variable is clearly imbalanced in the training set, with 16,355 non-default cases (77.88%) and 4,645 default cases (22.12%), so model evaluation should not rely on accuracy alone.

In [10]:

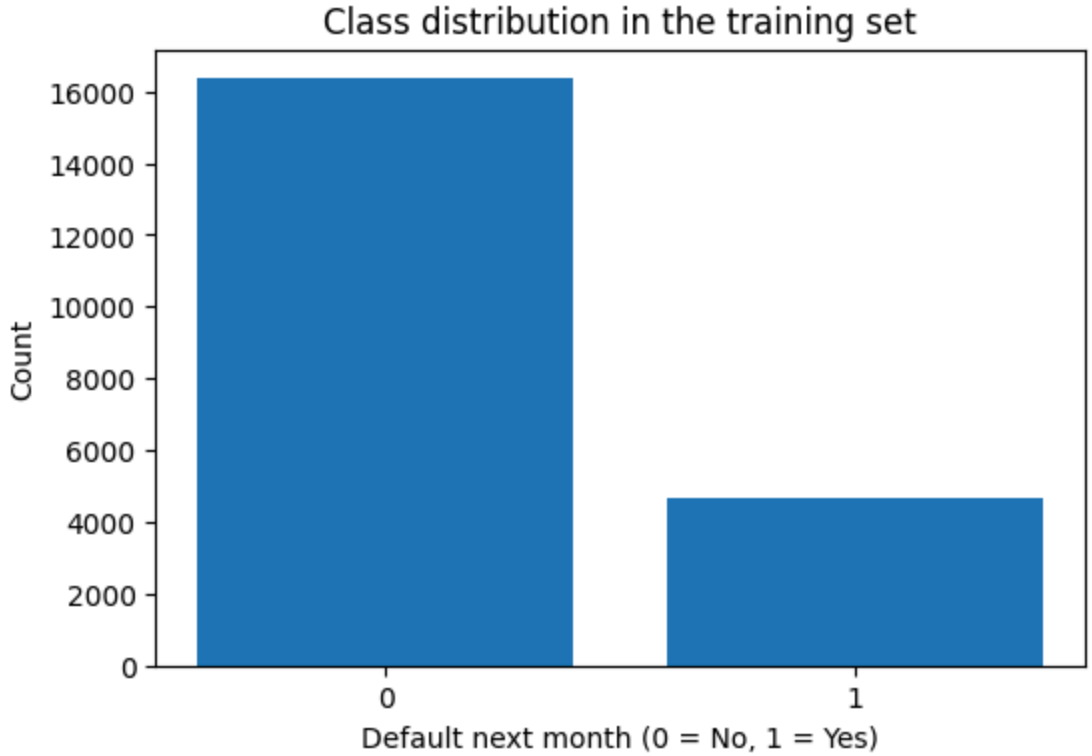
```python
selected_cols = ["LIMIT_BAL", "AGE", "PAY_0", "BILL_AMT1", "PAY_AMT1"]

print(train_df[selected_cols].describe().T)
```

```
              count           mean            std       min       25%  \
LIMIT_BAL  21000.0  167627.318095  129860.863421   10000.0  50000.00
AGE        21000.0      35.488476       9.224177      21.0     28.00
PAY_0      21000.0      -0.017429       1.125270      -2.0     -1.00
BILL_AMT1  21000.0   50978.436286   73097.849892 -154973.0   3517.75
PAY_AMT1   21000.0    5663.359190   16612.876887       0.0   1000.00

                 50%        75%        max
LIMIT_BAL  140000.0  240000.00  1000000.0
AGE            34.0      41.00       79.0
PAY_0           0.0       0.00        8.0
BILL_AMT1   22485.5   67061.25   964511.0
PAY_AMT1     2100.0    5001.00   873552.0
```

-- The selected features show very different scales and wide ranges. For example, `LIMIT_BAL` ranges from 10,000 to 1,000,000 and `PAY_AMT1` ranges from 0 to 873,552, which suggests that scaling will likely be important for some models.

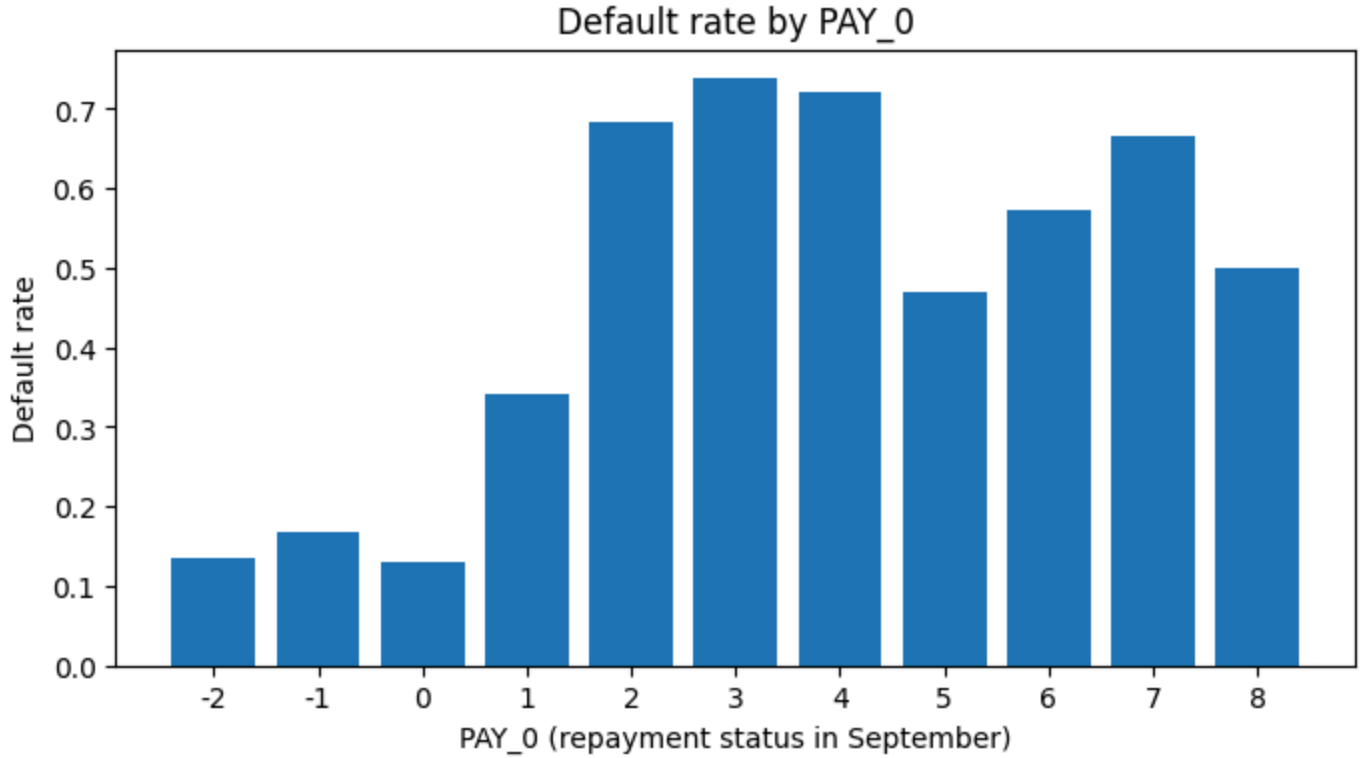
In [11]:

```python
#Visualization 1: class distribution bar chart
class_counts = train_df[target_col].value_counts().sort_index()

plt.figure(figsize=(6, 4))
plt.bar(class_counts.index.astype(str), class_counts.values)
plt.xlabel("Default next month (0 = No, 1 = Yes)")
plt.ylabel("Count")
plt.title("Class distribution in the training set")
plt.show()
```



-- The class distribution bar chart confirms a strong class imbalance, with the non-default class much larger than the default class.

In [12]:
```
#Visualization 2: default rate by repayment status
default_rate_by_pay0 = train_df.groupby("PAY_0")[target_col].mean().sort_index()

plt.figure(figsize=(8, 4))
plt.bar(default_rate_by_pay0.index.astype(str), default_rate_by_pay0.values)
plt.xlabel("PAY_0 (repayment status in September)")
plt.ylabel("Default rate")
plt.title("Default rate by PAY_0")
plt.show()

print(default_rate_by_pay0.round(3))
```




```
PAY_0
-2    0.134
-1    0.168
 0    0.129
 1    0.342
 2    0.684
 3    0.738
 4    0.722
 5    0.471
 6    0.571
 7    0.667
 8    0.500
Name: default.payment.next.month, dtype: float64
```

-- The default rate generally increases as `PAY_0` indicates worse recent repayment status. For example, around 0.129 at `PAY_0=0` versus `0.684 – 0.738` at `PAY_0=2` to 3, which shows that repayment status is likely a highly informative predictor.

Task 3:

The training dataset contains 21,000 rows and 25 columns, and the columns shown in the missing-value check have no missing values. The feature set includes both integer and float columns, and the variables are on very different numeric scales. I also observed that the target variable is imbalanced, which is about 78% non-default and 22% default, which means a naive model could appear to perform well if it

mostly predicts the majority class. In addition, the relationship between `PAY_0` and the default rate suggests that repayment history features are likely to be among the most useful predictors in the modeling stage.

Task 4:

Because this is an imbalanced binary classification problem, I will use ROC-AUC and F1-score as my main evaluation metrics, and I will also examine precision and recall. ROC-AUC is useful because it measures how well the model separates the two classes across thresholds, while F1-score balances precision and recall and is more informative than accuracy when the classes are imbalanced.

4. Feature engineering

---

rubric={points:1}

Your tasks:

1. **Perform feature engineering**: Create new features that are relevant to the problem and use this updated feature set in the following exercises. You may need to iterate between **feature engineering** and **preprocessing** to refine your features and improve model performance.

Solution_4

Points: 1

In [13]:
```
def add_credit_features(df):
    df_fe = df.copy()

    pay_status_cols = ["PAY_0", "PAY_2", "PAY_3", "PAY_4", "PAY_5", "PAY_6"]
    bill_cols = ["BILL_AMT1", "BILL_AMT2", "BILL_AMT3", "BILL_AMT4", "BILL_AMT5", "BILL_AMT6"]
    pay_amt_cols = ["PAY_AMT1", "PAY_AMT2", "PAY_AMT3", "PAY_AMT4", "PAY_AMT5", "PAY_AMT6"]

    #Repayment-status based features:
    df_fe["avg_pay_status"] = df_fe[pay_status_cols].mean(axis=1)
    df_fe["max_pay_status"] = df_fe[pay_status_cols].max(axis=1)
    df_fe["num_months_late"] = (df_fe[pay_status_cols] > 0).sum(axis=1)
    df_fe["any_month_late"] = (df_fe["num_months_late"] > 0).astype(int)
```

```python
    #Bill / payment amount aggregate features:
    df_fe["avg_bill_amt"] = df_fe[bill_cols].mean(axis=1)
    df_fe["avg_pay_amt"] = df_fe[pay_amt_cols].mean(axis=1)
    df_fe["total_bill_amt"] = df_fe[bill_cols].sum(axis=1)
    df_fe["total_pay_amt"] = df_fe[pay_amt_cols].sum(axis=1)

    #Average bill relative to credit limit:
    df_fe["mean_bill_to_limit"] = df_fe[bill_cols].mean(axis=1) / df_fe["LIMIT_BAL"]

    #Total payment relative to positive billed amount:
    positive_bill_sum = df_fe[bill_cols].clip(lower=0).sum(axis=1)
    df_fe["pay_to_bill_ratio_total"] = df_fe["total_pay_amt"] / (positive_bill_sum + 1)

    #Simple trend feature:
    df_fe["recent_bill_change"] = df_fe["BILL_AMT1"] - df_fe["BILL_AMT6"]

    return df_fe

#Apply to train and test:
X_train_fe = add_credit_features(X_train)
X_test_fe = add_credit_features(X_test)

print("Original X_train shape:", X_train.shape)
print("Feature-engineered X_train shape:", X_train_fe.shape)

new_cols = [c for c in X_train_fe.columns if c not in X_train.columns]
print("\nNew features created:")
print(new_cols)
```

```
Original X_train shape: (21000, 24)
Feature-engineered X_train shape: (21000, 35)

New features created:
['avg_pay_status', 'max_pay_status', 'num_months_late', 'any_month_late', 'avg_bill_amt', 'avg_pay_amt', 'total_bill_am
t', 'total_pay_amt', 'mean_bill_to_limit', 'pay_to_bill_ratio_total', 'recent_bill_change']
```

Task 1:

I performed feature engineering by creating new features that summarize repayment behavior, billing amounts, and payment behavior across multiple months. The new features are described in the comments.

These features are relevant because they capture patterns such as how often a client was late, how severe the repayment delay was, how large the bills are relative to the credit limit, and how much the client tends to pay relative to billed amounts.

After feature engineering, the number of training features increased from 24 to 35. I applied the same feature engineering function to both the training and test sets so the updated feature set can be used consistently.

5. Preprocessing and transformations

---

rubric={points:10}

Your tasks:

1. **Identify feature types**: Determine the different types of features in your dataset (e.g., numerical, categorical, ordinal, text) and specify the transformations you would apply to each type.
2. **Define a column transformer (if needed)**: Implement a `ColumnTransformer` to apply the appropriate preprocessing steps to each feature type.

Solution_5

Points: 10

In [14]:
```
from sklearn.compose import ColumnTransformer
from sklearn.pipeline import Pipeline
from sklearn.preprocessing import OneHotEncoder, StandardScaler
from sklearn.impute import SimpleImputer

id_cols = [col for col in ["ID"] if col in X_train_fe.columns]

categorical_cols = [col for col in ["SEX", "EDUCATION", "MARRIAGE"] if col in X_train_fe.columns]

numeric_cols = [col for col in X_train_fe.columns if col not in id_cols + categorical_cols]

print("ID columns (to drop):", id_cols)
print("Categorical columns (one-hot):", categorical_cols)
print("Number of numeric columns (scale):", len(numeric_cols))
print("\nFirst 10 numeric columns:", numeric_cols[:10])
```

```
ID columns (to drop): ['ID']
Categorical columns (one-hot): ['SEX', 'EDUCATION', 'MARRIAGE']
Number of numeric columns (scale): 31

First 10 numeric columns: ['LIMIT_BAL', 'AGE', 'PAY_0', 'PAY_2', 'PAY_3', 'PAY_4', 'PAY_5', 'PAY_6', 'BILL_AMT1', 'BILL
_AMT2']
```

In [15]:

```python
#Define:
numeric_transformer = Pipeline(steps=[
    ("imputer", SimpleImputer(strategy="median")),
    ("scaler", StandardScaler())
])

categorical_transformer = Pipeline(steps=[
    ("imputer", SimpleImputer(strategy="most_frequent")),
    ("onehot", OneHotEncoder(handle_unknown="ignore"))
])

preprocessor = ColumnTransformer(
    transformers=[
        ("num", numeric_transformer, numeric_cols),
        ("cat", categorical_transformer, categorical_cols),
        ("drop_id", "drop", id_cols)
    ],
    remainder="drop"
)

print(preprocessor)
```

```
ColumnTransformer(transformers=[('num',
                                 Pipeline(steps=[('imputer',
                                                  SimpleImputer(strategy='median')),
                                                 ('scaler', StandardScaler())]),
                                 ['LIMIT_BAL', 'AGE', 'PAY_0', 'PAY_2', 'PAY_3',
                                  'PAY_4', 'PAY_5', 'PAY_6', 'BILL_AMT1',
                                  'BILL_AMT2', 'BILL_AMT3', 'BILL_AMT4',
                                  'BILL_AMT5', 'BILL_AMT6', 'PAY_AMT1',
                                  'PAY_AMT2', 'PAY_AMT3', 'PAY_AMT4',
                                  'PAY_AMT5', 'PAY_AMT6', 'avg_pay_status',
                                  'max_pay_status', 'num_months_late',
                                  'any_month_late', 'avg_bill_amt',
                                  'avg_pay_amt', 'total_bill_amt',
                                  'total_pay_amt', 'mean_bill_to_limit',
                                  'pay_to_bill_ratio_total', ...]),
                                ('cat',
                                 Pipeline(steps=[('imputer',
                                                  SimpleImputer(strategy='most_frequent')),
                                                 ('onehot',
                                                  OneHotEncoder(handle_unknown='ignore'))]),
                                 ['SEX', 'EDUCATION', 'MARRIAGE']),
                                ('drop_id', 'drop', ['ID'])])
```

In [16]:

```python
X_train_preprocessed = preprocessor.fit_transform(X_train_fe)
X_test_preprocessed = preprocessor.transform(X_test_fe)

print("Preprocessed X_train shape:", X_train_preprocessed.shape)
print("Preprocessed X_test shape:", X_test_preprocessed.shape)
```

```
Preprocessed X_train shape: (21000, 44)
Preprocessed X_test shape: (9000, 44)
```

Small Conclude:

I implemented these transformations using a `ColumnTransformer`, which combines the numeric and categorical preprocessing pipelines into one object. Even though the dataset appears to have no missing values, I included imputation steps to make the preprocessing pipeline more robust.

6. Baseline model

---

rubric={points:2}

Your tasks:

1. **Establish a baseline**: Use one of `scikit-learn`'s baseline models (e.g., `DummyClassifier` or `DummyRegressor`, depending on your task) and report the results. This will serve as a reference point for evaluating the performance of your more advanced models.

Solution_6

Points: 2

In [17]:
```
from sklearn.dummy import DummyClassifier
from sklearn.model_selection import StratifiedKFold, cross_validate
from sklearn.metrics import make_scorer, precision_score, recall_score, f1_score

baseline_pipe = Pipeline(steps=[
    ("preprocessor", preprocessor),
    ("model", DummyClassifier(strategy="most_frequent"))
])

cv = StratifiedKFold(n_splits=5, shuffle=True, random_state=123)

scoring = {
    "accuracy": "accuracy",
    "precision": make_scorer(precision_score, zero_division=0),
    "recall": make_scorer(recall_score, zero_division=0),
    "f1": make_scorer(f1_score, zero_division=0),
    "roc_auc": "roc_auc"
}
```

In [18]:
```
baseline_cv = cross_validate(
    baseline_pipe,
    X_train_fe,
    y_train,
    cv=cv,
    scoring=scoring,
    return_train_score=False
)

baseline_cv
```

Out[18]:
```
{'fit_time': array([0.04498601, 0.0440619 , 0.04296684, 0.04190397, 0.04175305]),
 'score_time': array([0.01184297, 0.01129293, 0.01121092, 0.01118803, 0.01110983]),
 'test_accuracy': array([0.77880952, 0.77880952, 0.77880952, 0.77880952, 0.77880952]),
 'test_precision': array([0., 0., 0., 0., 0.]),
 'test_recall': array([0., 0., 0., 0., 0.]),
 'test_f1': array([0., 0., 0., 0., 0.]),
 'test_roc_auc': array([0.5, 0.5, 0.5, 0.5, 0.5])}
```

In [19]:
```python
baseline_summary = pd.DataFrame({
    "metric": ["accuracy", "precision", "recall", "f1", "roc_auc"],
    "mean_cv_score": [
        baseline_cv["test_accuracy"].mean(),
        baseline_cv["test_precision"].mean(),
        baseline_cv["test_recall"].mean(),
        baseline_cv["test_f1"].mean(),
        baseline_cv["test_roc_auc"].mean()
    ],
    "std_cv_score": [
        baseline_cv["test_accuracy"].std(),
        baseline_cv["test_precision"].std(),
        baseline_cv["test_recall"].std(),
        baseline_cv["test_f1"].std(),
        baseline_cv["test_roc_auc"].std()
    ]
})

baseline_summary.round(4)
```

Out[19]:

	metric	mean_cv_score	std_cv_score
0	accuracy	0.7788	0.0
1	precision	0.0000	0.0
2	recall	0.0000	0.0
3	f1	0.0000	0.0
4	roc_auc	0.5000	0.0

Comments on Baseline:

I established a baseline model using scikit-learn's DummyClassifier with strategy="most_frequent", which always predicts the majority class. I evaluated the baseline using 5-fold stratified cross-validation on the training set and reported the same metrics that I plan to use for later models.

The baseline achieved an accuracy of about 0.7788, which is high only because the dataset is imbalanced and the majority class is common. However, its precision, recall, and F1-score for the default class were all 0, because it never predicts the positive class. Its ROC-AUC was 0.50, which indicates no discriminative ability. This baseline is useful because it shows that a model must improve on more than just accuracy to be considered meaningful on this problem.

7. Linear models

---

rubric={points:10}

Your tasks:

1. **Train a linear model**: Use a linear model as your first real attempt at solving the problem.
2. **Tune hyperparameters**: Perform hyperparameter tuning to explore different values of the model's complexity parameter.
3. **Evaluate with cross-validation**: Report the cross-validation scores along with their standard deviation.
4. **Summarize findings**: Summarize your results, highlighting key observations from your experiments.

Solution_7

Points: 10

Type your answer here, replacing this text.

In [20]:
```
from sklearn.linear_model import LogisticRegression
from sklearn.model_selection import GridSearchCV

logreg_pipe = Pipeline(steps=[
    ("preprocessor", preprocessor),
    ("model", LogisticRegression(
        max_iter=2000,
        solver="liblinear",
        random_state=123
    ))
])
```

```
param_grid_logreg = {
    "model__C": [0.001, 0.01, 0.1, 1, 10, 100]
}
```

In [21]:
```
logreg_grid = GridSearchCV(
    estimator=logreg_pipe,
    param_grid=param_grid_logreg,
    scoring=scoring,
    refit="roc_auc",
    cv=cv,
    n_jobs=-1,
    return_train_score=True
)

logreg_grid.fit(X_train_fe, y_train)

print("Best parameters:", logreg_grid.best_params_)
print("Best CV ROC-AUC:", round(logreg_grid.best_score_, 4))
```

```
Best parameters: {'model__C': 100}
Best CV ROC-AUC: 0.7579
```

In [22]:
```
logreg_results = pd.DataFrame(logreg_grid.cv_results_)

cols_to_show = [
    "param_model__C",
    "mean_test_roc_auc", "std_test_roc_auc",
    "mean_test_f1", "std_test_f1",
    "mean_test_precision", "std_test_precision",
    "mean_test_recall", "std_test_recall",
    "mean_test_accuracy", "std_test_accuracy"
]

logreg_tuning_summary = logreg_results[cols_to_show].sort_values(by="mean_test_roc_auc", ascending=False)
logreg_tuning_summary.round(4)
```

Out[22]:

	param_model__C	mean_test_roc_auc	std_test_roc_auc	mean_test_f1	std_test_f1	mean_test_precision	std_test_precision	mean
5	100.000	0.7579	0.0093	0.4063	0.0098	0.6471	0.0128	
4	10.000	0.7579	0.0093	0.4063	0.0098	0.6471	0.0128	
3	1.000	0.7579	0.0092	0.4057	0.0097	0.6469	0.0123	
2	0.100	0.7576	0.0088	0.4048	0.0112	0.6488	0.0162	
1	0.010	0.7547	0.0077	0.4055	0.0084	0.6499	0.0140	
0	0.001	0.7439	0.0049	0.4059	0.0121	0.6477	0.0170	

In [23]:

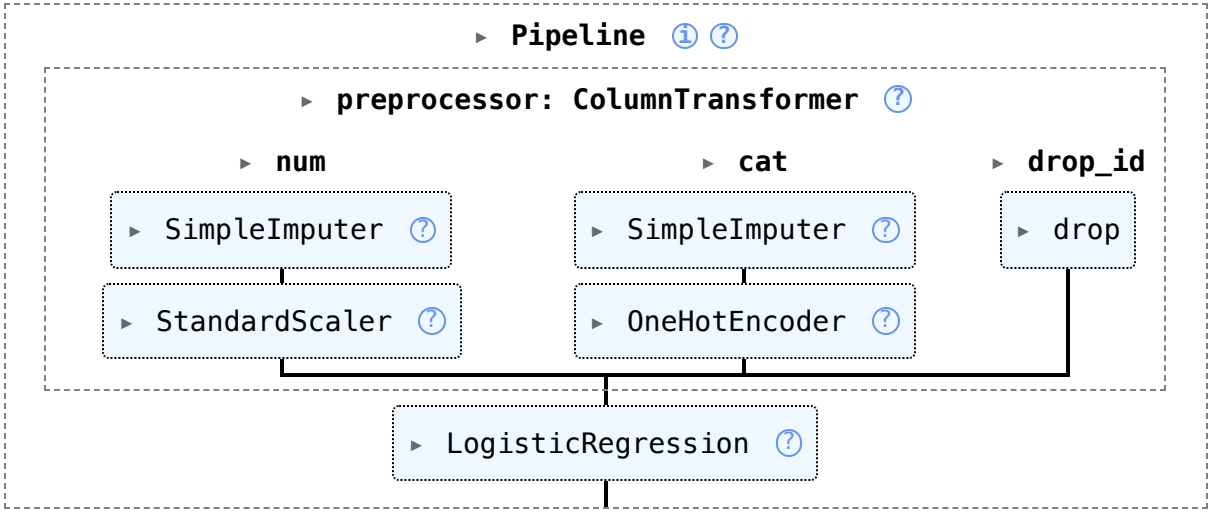
```
best_idx = logreg_grid.best_index_

linear_best_summary = pd.DataFrame({
    "metric": ["accuracy", "precision", "recall", "f1", "roc_auc"],
    "mean_cv_score": [
        logreg_results.loc[best_idx, "mean_test_accuracy"],
        logreg_results.loc[best_idx, "mean_test_precision"],
        logreg_results.loc[best_idx, "mean_test_recall"],
        logreg_results.loc[best_idx, "mean_test_f1"],
        logreg_results.loc[best_idx, "mean_test_roc_auc"]
    ],
    "std_cv_score": [
        logreg_results.loc[best_idx, "std_test_accuracy"],
        logreg_results.loc[best_idx, "std_test_precision"],
        logreg_results.loc[best_idx, "std_test_recall"],
        logreg_results.loc[best_idx, "std_test_f1"],
        logreg_results.loc[best_idx, "std_test_roc_auc"]
    ]
})

linear_best_summary.round(4)
```

Out[23]:

	metric	mean_cv_score	std_cv_score
0	accuracy	0.8086	0.0024
1	precision	0.6471	0.0128
2	recall	0.2962	0.0093
3	f1	0.4063	0.0098
4	roc_auc	0.7579	0.0093

In [24]:
```
best_linear_model = logreg_grid.best_estimator_
best_linear_model
```

Out[24]:



Summarize:

I trained a logistic regression model as my first real model for this binary classification problem. Logistic regression is a linear model for classification and is a strong benchmark because it is simple, interpretable, and works well with standardized numerical features and one-hot encoded categorical features.

I tuned the model's complexity parameter C using 5-fold stratified cross-validation and selected the best model based on ROC-AUC. The best hyperparameter value was C = 100, with a best cross-validation ROC-AUC of 0.7579.

For the best linear model, the cross-validation performance was approximately: accuracy = 0.8086 ± 0.0024, precision = 0.6471 ± 0.0128, recall = 0.2962 ± 0.0093, F1-score = 0.4063 ± 0.0098, and ROC-AUC = 0.7579 ± 0.0093. Compared with the DummyClassifier baseline, the logistic regression model substantially improved ROC-AUC, precision, recall, and F1-score, showing that it learned meaningful patterns rather than simply predicting the majority class.

From the tuning results, I observed that performance was fairly stable across the tested values of C, with the best ROC-AUC values for C = 10 and C = 100 being essentially the same. This suggests that the model is not highly sensitive to regularization within this range. Overall, logistic regression provides a strong and efficient linear benchmark for comparison with more complex models in the next section.

8. Different models

rubric={points:12}

Your tasks:

1. **Experiment with additional models**: Train at least **three models** other than a linear model. Ensure that **at least one** of these models is a **tree-based ensemble model** (e.g., Random Forest, Gradient Boosting, or XGBoost).

2. **Compare and interpret results**: Summarize your findings in terms of **overfitting/underfitting** behavior and **fit/score times** for each model. Reflect on your results. Were you able to **outperform the linear model**?

Solution_8

Points: 12

Type your answer here, replacing this text.

In [25]:
```
from sklearn.tree import DecisionTreeClassifier
from sklearn.neighbors import KNeighborsClassifier
from sklearn.ensemble import RandomForestClassifier

nonlinear_models = {
    "DecisionTree": Pipeline(steps=[
        ("preprocessor", preprocessor),
        ("model", DecisionTreeClassifier(random_state=123))
    ]),

    "KNN": Pipeline(steps=[
        ("preprocessor", preprocessor),
        ("model", KNeighborsClassifier(n_neighbors=15))
    ]),

    "RandomForest": Pipeline(steps=[
        ("preprocessor", preprocessor),
        ("model", RandomForestClassifier(
            n_estimators=300,
            random_state=123,
            n_jobs=-1
        ))
    ])
}
```

```
nonlinear_models
```

```
Out[25]: {'DecisionTree': Pipeline(steps=[('preprocessor',
                          ColumnTransformer(transformers=[('num',
                                                           Pipeline(steps=[('imputer',
                                                                            SimpleImputer(strategy='median')),
                                                                           ('scaler',
                                                                            StandardScaler())]),
                                                           ['LIMIT_BAL', 'AGE', 'PAY_0',
                                                            'PAY_2', 'PAY_3', 'PAY_4',
                                                            'PAY_5', 'PAY_6',
                                                            'BILL_AMT1', 'BILL_AMT2',
                                                            'BILL_AMT3', 'BILL_AMT4',
                                                            'BILL_AMT5', 'BILL_AMT6',
                                                            'PAY_AMT1', 'PAY_AMT2',
                                                            'PAY_AMT3', 'PAY_AMT4',
                                                            'PAY_...
                                                            'avg_bill_amt',
                                                            'avg_pay_amt',
                                                            'total_bill_amt',
                                                            'total_pay_amt',
                                                            'mean_bill_to_limit',
                                                            'pay_to_bill_ratio_total', ...]),
                                                          ('cat',
                                                           Pipeline(steps=[('imputer',
                                                                            SimpleImputer(strategy='most_frequent')),
                                                                           ('onehot',
                                                                            OneHotEncoder(handle_unknown='ignore'))]),
                                                           ['SEX', 'EDUCATION',
                                                            'MARRIAGE']),
                                                          ('drop_id', 'drop', ['ID'])])),
                         ('model', DecisionTreeClassifier(random_state=123))]),
 'KNN': Pipeline(steps=[('preprocessor',
                          ColumnTransformer(transformers=[('num',
                                                           Pipeline(steps=[('imputer',
                                                                            SimpleImputer(strategy='median')),
                                                                           ('scaler',
                                                                            StandardScaler())]),
                                                           ['LIMIT_BAL', 'AGE', 'PAY_0',
                                                            'PAY_2', 'PAY_3', 'PAY_4',
                                                            'PAY_5', 'PAY_6',
                                                            'BILL_AMT1', 'BILL_AMT2',
                                                            'BILL_AMT3', 'BILL_AMT4',
                                                            'BILL_AMT5', 'BILL_AMT6',
                                                            'PAY_AMT1', 'PAY_AMT2',
                                                            'PAY_AMT3', 'PAY_AMT4',
                                                            'PAY_...
                                                            'avg_bill_amt',
                                                            'avg_pay_amt',
```

```
                                                 'total_bill_amt',
                                                 'total_pay_amt',
                                                 'mean_bill_to_limit',
                                                 'pay_to_bill_ratio_total', ...]),
                                                ('cat',
                                                 Pipeline(steps=[('imputer',
                                                                  SimpleImputer(strategy='most_frequent')),
                                                                 ('onehot',
                                                                  OneHotEncoder(handle_unknown='ignore'))]),
                                                 ['SEX', 'EDUCATION',
                                                  'MARRIAGE']),
                                                ('drop_id', 'drop', ['ID'])])),
                                 ('model', KNeighborsClassifier(n_neighbors=15))]),
 'RandomForest': Pipeline(steps=[('preprocessor',
                                  ColumnTransformer(transformers=[('num',
                                                 Pipeline(steps=[('imputer',
                                                                  SimpleImputer(strategy='median')),
                                                                 ('scaler',
                                                                  StandardScaler())]),
                                                 ['LIMIT_BAL', 'AGE', 'PAY_0',
                                                  'PAY_2', 'PAY_3', 'PAY_4',
                                                  'PAY_5', 'PAY_6',
                                                  'BILL_AMT1', 'BILL_AMT2',
                                                  'BILL_AMT3', 'BILL_AMT4',
                                                  'BILL_AMT5', 'BILL_AMT6',
                                                  'PAY_AMT1', 'PAY_AMT2',
                                                  'PAY_AMT3', 'PAY_AMT4',
                                                  'PAY_...
                                                  'total_bill_amt',
                                                  'total_pay_amt',
                                                  'mean_bill_to_limit',
                                                  'pay_to_bill_ratio_total', ...]),
                                                ('cat',
                                                 Pipeline(steps=[('imputer',
                                                                  SimpleImputer(strategy='most_frequent')),
                                                                 ('onehot',
                                                                  OneHotEncoder(handle_unknown='ignore'))]),
                                                 ['SEX', 'EDUCATION',
                                                  'MARRIAGE']),
                                                ('drop_id', 'drop', ['ID'])])),
                                 ('model',
                                  RandomForestClassifier(n_estimators=300, n_jobs=-1,
                                                         random_state=123))])}
```

In [26]:

```python
from sklearn.model_selection import cross_validate

nonlinear_cv_results = {}
nonlinear_rows = []
```

```python
for model_name, model_pipe in nonlinear_models.items():
    cv_result = cross_validate(
        model_pipe,
        X_train_fe,
        y_train,
        cv=cv,
        scoring=scoring,
        return_train_score=True
    )

    nonlinear_cv_results[model_name] = cv_result

    nonlinear_rows.append({
        "model": model_name,

        #timing
        "fit_time_mean": cv_result["fit_time"].mean(),
        "score_time_mean": cv_result["score_time"].mean(),

        #validation metrics
        "test_accuracy_mean": cv_result["test_accuracy"].mean(),
        "test_precision_mean": cv_result["test_precision"].mean(),
        "test_recall_mean": cv_result["test_recall"].mean(),
        "test_f1_mean": cv_result["test_f1"].mean(),
        "test_roc_auc_mean": cv_result["test_roc_auc"].mean(),

        #train metrics
        "train_accuracy_mean": cv_result["train_accuracy"].mean(),
        "train_f1_mean": cv_result["train_f1"].mean(),
        "train_roc_auc_mean": cv_result["train_roc_auc"].mean(),

        #std of validation metrics
        "test_f1_std": cv_result["test_f1"].std(),
        "test_roc_auc_std": cv_result["test_roc_auc"].std()
    })

nonlinear_summary = pd.DataFrame(nonlinear_rows)
nonlinear_summary
```

Out[26]:

	model	fit_time_mean	score_time_mean	test_accuracy_mean	test_precision_mean	test_recall_mean	test_f1_mean	test_r
0	DecisionTree	0.430198	0.012819	0.722333	0.380752	0.408611	0.394154	
1	KNN	0.043935	0.175731	0.808571	0.622949	0.340797	0.440413	
2	RandomForest	2.017447	0.116984	0.814095	0.635916	0.373089	0.470246	

In [27]:
```
nonlinear_summary["roc_auc_gap_train_minus_test"] = (
    nonlinear_summary["train_roc_auc_mean"] - nonlinear_summary["test_roc_auc_mean"]
)

nonlinear_summary["f1_gap_train_minus_test"] = (
    nonlinear_summary["train_f1_mean"] - nonlinear_summary["test_f1_mean"]
)

nonlinear_summary_sorted = nonlinear_summary.sort_values(
    by="test_roc_auc_mean", ascending=False
)

nonlinear_summary_sorted.round(4)
```

Out[27]:

	model	fit_time_mean	score_time_mean	test_accuracy_mean	test_precision_mean	test_recall_mean	test_f1_mean	test_r
2	RandomForest	2.0174	0.1170	0.8141	0.6359	0.3731	0.4702	
1	KNN	0.0439	0.1757	0.8086	0.6229	0.3408	0.4404	
0	DecisionTree	0.4302	0.0128	0.7223	0.3808	0.4086	0.3942	

In [28]:
```
#Extract best linear model CV metrics from Q7 table:
linear_metrics = linear_best_summary.set_index("metric")["mean_cv_score"].to_dict()

linear_comparison = pd.DataFrame({
    "model": ["LogisticRegression (best linear)"],
    "test_accuracy_mean": [linear_metrics["accuracy"]],
    "test_precision_mean": [linear_metrics["precision"]],
    "test_recall_mean": [linear_metrics["recall"]],
    "test_f1_mean": [linear_metrics["f1"]],
    "test_roc_auc_mean": [linear_metrics["roc_auc"]]
})

comparison_table = pd.concat([
    linear_comparison,
    nonlinear_summary_sorted[[
        "model", "test_accuracy_mean", "test_precision_mean",
        "test_recall_mean", "test_f1_mean", "test_roc_auc_mean"
    ]]
], ignore_index=True)

comparison_table.round(4)
```

Out[28]:

	model	test_accuracy_mean	test_precision_mean	test_recall_mean	test_f1_mean	test_roc_auc_mean
0	LogisticRegression (best linear)	0.8086	0.6471	0.2962	0.4063	0.7579
1	RandomForest	0.8141	0.6359	0.3731	0.4702	0.7691
2	KNN	0.8086	0.6229	0.3408	0.4404	0.7399
3	DecisionTree	0.7223	0.3808	0.4086	0.3942	0.6103

In [29]:

```
best_nonlinear_name = nonlinear_summary_sorted.iloc[0]["model"]
print("Best non-linear model by CV ROC-AUC:", best_nonlinear_name)
```

```
Best non-linear model by CV ROC-AUC: RandomForest
```

Summarize:

I experimented with three non-linear models: Decision Tree, KNN, and Random Forest. I evaluated all models using the same 5-fold stratified cross-validation setup and the same metrics as in previous sections so that the comparison with the linear model would be fair.

Based on the cross-validation results, the best-performing non-linear model in terms of ROC-AUC was Random Forest, with a mean ROC-AUC of 0.7691 and mean F1-score of 0.4702. The KNN model achieved a mean ROC-AUC of 0.7399 and mean F1-score of 0.4404, while the Decision Tree model achieved a mean ROC-AUC of 0.6103 and mean F1-score of 0.3942. In comparison, the best linear model had a mean ROC-AUC of 0.7579 and mean F1-score of 0.4063.

In terms of overfitting and underfitting behavior, the Decision Tree showed strong overfitting. Its training accuracy was extremely high, about 0.9996, while its validation accuracy dropped to 0.7223, and its validation ROC-AUC was much lower than the other models. This indicates that the single tree fit the training data too closely and generalized poorly. The KNN model showed much less overfitting and produced reasonable performance, but it did not outperform logistic regression on ROC-AUC. The Random Forest model also showed some overfitting (very high training accuracy), but it generalized much better than the single Decision Tree and achieved the best validation performance overall.

Regarding computation time, KNN had the fastest fit time of 0.0466s but the slowest score time of 0.1992s because prediction requires distance calculations. Random Forest had the slowest fit time of about 2.0236s but a moderate score time of 0.1161s. Decision Tree had a moderate fit time of 0.4316s and the fastest score time of 0.0138s.

Overall, Random Forest improved accuracy (0.8141 vs 0.8086), recall (0.3731 vs 0.2962), F1-score (0.4702 vs 0.4063), and ROC-AUC (0.7691 vs 0.7579) compared with logistic regression, although its precision (0.6359) was slightly lower than logistic regression (0.6471). This comparison highlights useful trade-offs between predictive performance, overfitting risk, and computational cost.

9. Feature selection

rubric={points:2}

Your tasks:

1. **Perform feature selection**: Attempt to select relevant features using methods such as `RFECV` or forward selection.
2. **Evaluate the impact** Compare the model performance before and after feature selection. Do the results improve with feature selection?
3. **Summarize findings** Summarize your observations and decide whether to **keep feature selection** in your pipeline. If it improves results, retain it for the next exercises; otherwise, you may choose to omit it.

Solution_9

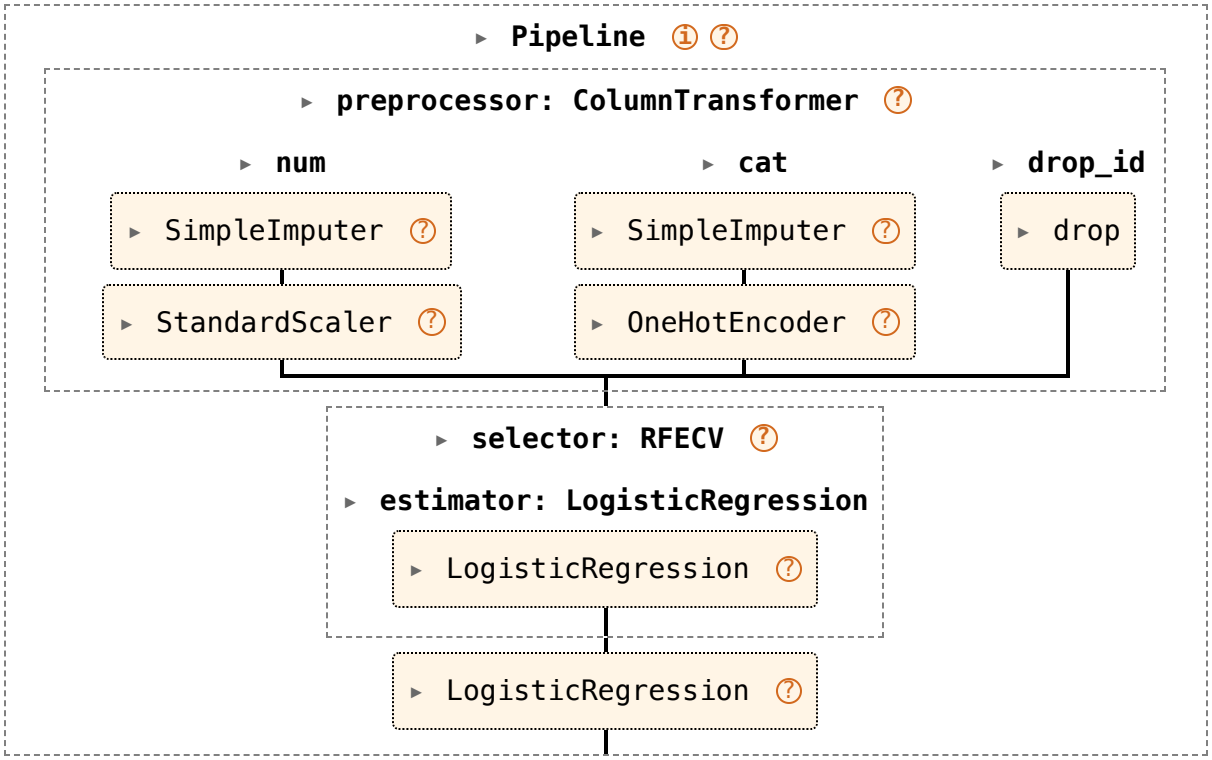
Points: 2

Type your answer here, replacing this text.

In [30]:
```
from sklearn.feature_selection import RFECV

rfecv_estimator = LogisticRegression(
    C=100,
    max_iter=2000,
    solver="liblinear",
    random_state=123
)

#RFECV:
rfecv = RFECV(
    estimator=rfecv_estimator,
    step=2,
    min_features_to_select=10,
    cv=StratifiedKFold(n_splits=3, shuffle=True, random_state=123),
    scoring="roc_auc",
```

```python
    n_jobs=-1
)

logreg_fs_pipe = Pipeline(steps=[
    ("preprocessor", preprocessor),
    ("selector", rfecv),
    ("model", LogisticRegression(
        C=100,
        max_iter=2000,
        solver="liblinear",
        random_state=123
    ))
])

logreg_fs_pipe
```

Out[30]:



In [31]:

```python
logreg_fs_cv = cross_validate(
    logreg_fs_pipe,
    X_train_fe,
    y_train,
    cv=cv,
    scoring=scoring,
    return_train_score=True
)
```

```
logreg_fs_cv
```

Out[31]:

```
{'fit_time': array([2.68234086, 2.11339784, 2.59081602, 2.55877399, 2.77803421]),
 'score_time': array([0.0132761 , 0.01335406, 0.01349998, 0.01858473, 0.01305079]),
 'test_accuracy': array([0.81071429, 0.80738095, 0.80595238, 0.81166667, 0.80619048]),
 'train_accuracy': array([0.8075    , 0.80964286, 0.80970238, 0.80821429, 0.81083333]),
 'test_precision': array([0.65509259, 0.6369863 , 0.64179104, 0.66197183, 0.63218391]),
 'train_precision': array([0.64465786, 0.6509324 , 0.65219941, 0.64546525, 0.65389016]),
 'test_recall': array([0.30462863, 0.30032293, 0.27771798, 0.30355221, 0.29601722]),
 'train_recall': array([0.28902045, 0.30059203, 0.2992465 , 0.2949408 , 0.30758881]),
 'test_f1': array([0.41587068, 0.40819312, 0.38767844, 0.41623616, 0.40322581]),
 'train_f1': array([0.39910814, 0.41126657, 0.41025641, 0.40487625, 0.41837482]),
 'test_roc_auc': array([0.77151215, 0.75127972, 0.7620963 , 0.75926225, 0.74423046]),
 'train_roc_auc': array([0.75804931, 0.7635233 , 0.76032376, 0.76200706, 0.76376312])}
```

In [32]:

```
fs_summary = pd.DataFrame({
    "metric": ["accuracy", "precision", "recall", "f1", "roc_auc"],
    "mean_cv_score": [
        logreg_fs_cv["test_accuracy"].mean(),
        logreg_fs_cv["test_precision"].mean(),
        logreg_fs_cv["test_recall"].mean(),
        logreg_fs_cv["test_f1"].mean(),
        logreg_fs_cv["test_roc_auc"].mean()
    ],
    "std_cv_score": [
        logreg_fs_cv["test_accuracy"].std(),
        logreg_fs_cv["test_precision"].std(),
        logreg_fs_cv["test_recall"].std(),
        logreg_fs_cv["test_f1"].std(),
        logreg_fs_cv["test_roc_auc"].std()
    ]
})

fs_summary.round(4)
```

Out[32]:

	metric	mean_cv_score	std_cv_score
0	accuracy	0.8084	0.0024
1	precision	0.6456	0.0112
2	recall	0.2964	0.0098
3	f1	0.4062	0.0105
4	roc_auc	0.7577	0.0093

In [33]:
```
#Q7 best logistic regression results are stored in linear_best_summary:
before_fs_summary = linear_best_summary.copy()
before_fs_summary["version"] = "Before feature selection"

after_fs_summary = fs_summary.copy()
after_fs_summary["version"] = "After feature selection"

feature_selection_comparison = pd.concat(
    [before_fs_summary, after_fs_summary],
    ignore_index=True
)

feature_selection_comparison = feature_selection_comparison[["version", "metric", "mean_cv_score", "std_cv_score"]]
feature_selection_comparison.round(4)
```

Out[33]:

	version	metric	mean_cv_score	std_cv_score
0	Before feature selection	accuracy	0.8086	0.0024
1	Before feature selection	precision	0.6471	0.0128
2	Before feature selection	recall	0.2962	0.0093
3	Before feature selection	f1	0.4063	0.0098
4	Before feature selection	roc_auc	0.7579	0.0093
5	After feature selection	accuracy	0.8084	0.0024
6	After feature selection	precision	0.6456	0.0112
7	After feature selection	recall	0.2964	0.0098
8	After feature selection	f1	0.4062	0.0105
9	After feature selection	roc_auc	0.7577	0.0093

In [34]:
```
#Fit once on the full training set to inspect what RFECV selected:
logreg_fs_pipe.fit(X_train_fe, y_train)

fitted_preprocessor = logreg_fs_pipe.named_steps["preprocessor"]
fitted_selector = logreg_fs_pipe.named_steps["selector"]

feature_names_after_preprocessing = fitted_preprocessor.get_feature_names_out()

selected_mask = fitted_selector.support_
selected_feature_names = feature_names_after_preprocessing[selected_mask]

print("Number of features after preprocessing:", len(feature_names_after_preprocessing))
```

```python
print("Number of selected features:", fitted_selector.n_features_)
print("\nSelected features:")
print(selected_feature_names)
```

```
Number of features after preprocessing: 44
Number of selected features: 40

Selected features:
['num__LIMIT_BAL' 'num__AGE' 'num__PAY_0' 'num__PAY_2' 'num__PAY_3'
 'num__PAY_4' 'num__PAY_5' 'num__PAY_6' 'num__BILL_AMT1' 'num__BILL_AMT2'
 'num__BILL_AMT3' 'num__BILL_AMT4' 'num__BILL_AMT5' 'num__BILL_AMT6'
 'num__PAY_AMT1' 'num__PAY_AMT2' 'num__PAY_AMT3' 'num__PAY_AMT6'
 'num__avg_pay_status' 'num__num_months_late' 'num__any_month_late'
 'num__avg_bill_amt' 'num__avg_pay_amt' 'num__total_bill_amt'
 'num__total_pay_amt' 'num__mean_bill_to_limit' 'num__recent_bill_change'
 'cat__SEX_1' 'cat__SEX_2' 'cat__EDUCATION_0' 'cat__EDUCATION_1'
 'cat__EDUCATION_2' 'cat__EDUCATION_3' 'cat__EDUCATION_4'
 'cat__EDUCATION_5' 'cat__EDUCATION_6' 'cat__MARRIAGE_0' 'cat__MARRIAGE_1'
 'cat__MARRIAGE_2' 'cat__MARRIAGE_3']
```

Summarize:

I performed feature selection using RFECV with logistic regression as the estimator and ROC-AUC as the selection metric. I placed RFECV inside a pipeline with the preprocessor so that feature selection was performed only on the training folds during cross-validation, which helps avoid data leakage.

RFECV selected 40 features out of 44 preprocessed features. I then compared the logistic regression model performance before and after feature selection using the same 5-fold stratified cross-validation setup. Before feature selection, the logistic regression model had ROC-AUC = 0.7579 and F1-score = 0.4063. After feature selection, the model achieved ROC-AUC = 0.7577 and F1-score = 0.4062.

The results stayed almost the same, with a very small decrease in ROC-AUC and F1-score after feature selection. Accuracy also changed only slightly about 0.8086 to 0.8084, and precision/recall remained nearly unchanged. Based on this comparison, I will NOT keep feature selection in the pipeline for the next sections because it did not provide a meaningful performance improvement and the original pipeline is simpler.

Even though the performance change was negligible, this experiment was still useful because it showed that the logistic regression model was already using a feature set that was not heavily hurt by a small amount of redundancy.

10. Hyperparameter optimization

rubric={points:10}

Your tasks:

1. **Optimize hyperparameters**: Attempt to optimize hyperparameters for the models you have tried so far. In at least **one case**, tune **multiple hyperparameters** for a single model.
2. **Use suitable optimization methods**: You may use any of the following approaches for hyperparameter optimization:
   - `GridSearchCV`
   - `RandomizedSearchCV`
   - Bayesian optimization with scikit-optimize
3. **Summarize your results**: Report and compare the optimized results across models. Discuss whether hyperparameter optimization led to performance improvements.

Solution_10

Points: 10

Type your answer here, replacing this text.

In [35]:
```
from sklearn.model_selection import RandomizedSearchCV
cv_q10 = StratifiedKFold(n_splits=3, shuffle=True, random_state=123)
# ----------------------------
# 1) Logistic Regression search
# ----------------------------
logreg_tune_pipe = Pipeline(steps=[
    ("preprocessor", preprocessor),
    ("model", LogisticRegression(
        max_iter=2000,
        solver="liblinear",
        random_state=123
    ))
])

logreg_tune_grid = {
    "model__C": [0.01, 0.1, 1, 10, 100],
    "model__class_weight": [None, "balanced"]
}
```

```python
logreg_search = GridSearchCV(
    estimator=logreg_tune_pipe,
    param_grid=logreg_tune_grid,
    scoring=scoring,
    refit="roc_auc",
    cv=cv_q10,
    n_jobs=-1,
    return_train_score=True
)

# ----------------------------
# 2) KNN search
# ----------------------------
knn_tune_pipe = Pipeline(steps=[
    ("preprocessor", preprocessor),
    ("model", KNeighborsClassifier())
])

knn_tune_grid = {
    "model__n_neighbors": [10, 15, 25],
    "model__weights": ["uniform", "distance"],
    "model__p": [1, 2]
}

knn_search = GridSearchCV(
    estimator=knn_tune_pipe,
    param_grid=knn_tune_grid,
    scoring=scoring,
    refit="roc_auc",
    cv=cv_q10,
    n_jobs=-1,
    return_train_score=True
)

# ----------------------------
# 3) Random Forest search
# ----------------------------
rf_tune_pipe = Pipeline(steps=[
    ("preprocessor", preprocessor),
    ("model", RandomForestClassifier(random_state=123, n_jobs=-1))
])

rf_param_dist = {
    "model__n_estimators": [200, 300, 500],
    "model__max_depth": [None, 10, 20, 30],
    "model__min_samples_split": [2, 5, 10],
    "model__min_samples_leaf": [1, 2, 4],
    "model__max_features": ["sqrt", "log2", None],
```

```
    "model__class_weight": [None, "balanced"]
}

rf_search = RandomizedSearchCV(
    estimator=rf_tune_pipe,
    param_distributions=rf_param_dist,
    n_iter=6,
    scoring=scoring,
    refit="roc_auc",
    cv=cv_q10,
    random_state=123,
    n_jobs=-1,
    return_train_score=True
)

print("Search objects created.")
```

```
Search objects created.
```

In [36]:
```
logreg_search.fit(X_train_fe, y_train)

print("LogisticRegression best params:", logreg_search.best_params_)
print("LogisticRegression best CV ROC-AUC:", round(logreg_search.best_score_, 4))
```

```
LogisticRegression best params: {'model__C': 100, 'model__class_weight': 'balanced'}
LogisticRegression best CV ROC-AUC: 0.7596
```

In [37]:
```
knn_search.fit(X_train_fe, y_train)

print("KNN best params:", knn_search.best_params_)
print("KNN best CV ROC-AUC:", round(knn_search.best_score_, 4))
```

```
KNN best params: {'model__n_neighbors': 25, 'model__p': 1, 'model__weights': 'uniform'}
KNN best CV ROC-AUC: 0.7581
```

In [38]:
```
rf_search.fit(X_train_fe, y_train)

print("RandomForest best params:", rf_search.best_params_)
print("RandomForest best CV ROC-AUC:", round(rf_search.best_score_, 4))
```

```
RandomForest best params: {'model__n_estimators': 500, 'model__min_samples_split': 10, 'model__min_samples_leaf': 2, 'm
odel__max_features': 'sqrt', 'model__max_depth': 10, 'model__class_weight': None}
RandomForest best CV ROC-AUC: 0.7822
```

In [39]:
```
def summarize_search(search_obj, model_name):
    results = pd.DataFrame(search_obj.cv_results_)
    best_idx = search_obj.best_index_

    row = {
```

```
        "model": model_name,
        "best_params": search_obj.best_params_,
        "fit_time_mean_best": results.loc[best_idx, "mean_fit_time"],
        "score_time_mean_best": results.loc[best_idx, "mean_score_time"],

        "cv_accuracy_mean": results.loc[best_idx, "mean_test_accuracy"],
        "cv_precision_mean": results.loc[best_idx, "mean_test_precision"],
        "cv_recall_mean": results.loc[best_idx, "mean_test_recall"],
        "cv_f1_mean": results.loc[best_idx, "mean_test_f1"],
        "cv_roc_auc_mean": results.loc[best_idx, "mean_test_roc_auc"],

        "cv_accuracy_std": results.loc[best_idx, "std_test_accuracy"],
        "cv_precision_std": results.loc[best_idx, "std_test_precision"],
        "cv_recall_std": results.loc[best_idx, "std_test_recall"],
        "cv_f1_std": results.loc[best_idx, "std_test_f1"],
        "cv_roc_auc_std": results.loc[best_idx, "std_test_roc_auc"],
    }
    return row

optimized_rows = [
    summarize_search(logreg_search, "LogisticRegression (optimized)"),
    summarize_search(knn_search, "KNN (optimized)"),
    summarize_search(rf_search, "RandomForest (optimized)")
]

optimized_summary = pd.DataFrame(optimized_rows).sort_values(
    by="cv_roc_auc_mean", ascending=False
)

optimized_summary.round(4)
```

Out[39]:

	model	best_params	fit_time_mean_best	score_time_mean_best	cv_accuracy_mean	cv_precision_mean	cv_re
2	RandomForest (optimized)	{'model__n_estimators': 500, 'model__min_sampl...	15.9984	0.6272	0.8185	0.6677	
0	LogisticRegression (optimized)	{'model__C': 100, 'model__class_weight': 'bala...	0.7288	0.0240	0.7478	0.4500	
1	KNN (optimized)	{'model__n_neighbors': 25, 'model__p': 1, 'mod...	0.0593	6.9822	0.8116	0.6370	

In [40]:

```
linear_metrics_before = linear_best_summary.set_index("metric")["mean_cv_score"].to_dict()

nonlinear_metrics_before = nonlinear_summary.set_index("model")
```

```
before_rows = [
    {
        "model": "LogisticRegression",
        "version": "Before optimization",
        "cv_accuracy_mean": linear_metrics_before["accuracy"],
        "cv_precision_mean": linear_metrics_before["precision"],
        "cv_recall_mean": linear_metrics_before["recall"],
        "cv_f1_mean": linear_metrics_before["f1"],
        "cv_roc_auc_mean": linear_metrics_before["roc_auc"]
    },
    {
        "model": "KNN",
        "version": "Before optimization",
        "cv_accuracy_mean": nonlinear_metrics_before.loc["KNN", "test_accuracy_mean"],
        "cv_precision_mean": nonlinear_metrics_before.loc["KNN", "test_precision_mean"],
        "cv_recall_mean": nonlinear_metrics_before.loc["KNN", "test_recall_mean"],
        "cv_f1_mean": nonlinear_metrics_before.loc["KNN", "test_f1_mean"],
        "cv_roc_auc_mean": nonlinear_metrics_before.loc["KNN", "test_roc_auc_mean"]
    },
    {
        "model": "RandomForest",
        "version": "Before optimization",
        "cv_accuracy_mean": nonlinear_metrics_before.loc["RandomForest", "test_accuracy_mean"],
        "cv_precision_mean": nonlinear_metrics_before.loc["RandomForest", "test_precision_mean"],
        "cv_recall_mean": nonlinear_metrics_before.loc["RandomForest", "test_recall_mean"],
        "cv_f1_mean": nonlinear_metrics_before.loc["RandomForest", "test_f1_mean"],
        "cv_roc_auc_mean": nonlinear_metrics_before.loc["RandomForest", "test_roc_auc_mean"]
    }
]

before_optimization_summary = pd.DataFrame(before_rows)

after_optimization_summary = optimized_summary.copy()
after_optimization_summary["model"] = after_optimization_summary["model"].str.replace(" \\(optimized\\)", "", regex=Tru
after_optimization_summary["version"] = "After optimization"

compare_opt = pd.concat([
    before_optimization_summary[["model", "version", "cv_accuracy_mean", "cv_precision_mean", "cv_recall_mean", "cv_f1_
    after_optimization_summary[["model", "version", "cv_accuracy_mean", "cv_precision_mean", "cv_recall_mean", "cv_f1_m
], ignore_index=True)

compare_opt = compare_opt.sort_values(["model", "version"])
compare_opt.round(4)
```

Out[40]:

	model	version	cv_accuracy_mean	cv_precision_mean	cv_recall_mean	cv_f1_mean	cv_roc_auc_mean
5	KNN	After optimization	0.8116	0.6370	0.3449	0.4473	0.7581
1	KNN	Before optimization	0.8086	0.6229	0.3408	0.4404	0.7399
4	LogisticRegression	After optimization	0.7478	0.4500	0.6243	0.5229	0.7596
0	LogisticRegression	Before optimization	0.8086	0.6471	0.2962	0.4063	0.7579
3	RandomForest	After optimization	0.8185	0.6677	0.3574	0.4654	0.7822
2	RandomForest	Before optimization	0.8141	0.6359	0.3731	0.4702	0.7691

In [41]:

```
search_dict = {
    "LogisticRegression": logreg_search,
    "KNN": knn_search,
    "RandomForest": rf_search
}

best_model_name = max(search_dict, key=lambda k: search_dict[k].best_score_)
best_search = search_dict[best_model_name]
best_optimized_model = best_search.best_estimator_

print("Best optimized model by CV ROC-AUC:", best_model_name)
print("Best optimized CV ROC-AUC:", round(best_search.best_score_, 4))
print("Best params:", best_search.best_params_)
```

```
Best optimized model by CV ROC-AUC: RandomForest
Best optimized CV ROC-AUC: 0.7822
Best params: {'model__n_estimators': 500, 'model__min_samples_split': 10, 'model__min_samples_leaf': 2, 'model__max_fea
tures': 'sqrt', 'model__max_depth': 10, 'model__class_weight': None}
```

Summarize:

I performed hyperparameter optimization for several models I had already tried, including Logistic Regression, KNN, and Random Forest. I used GridSearchCV for Logistic Regression and KNN, and RandomizedSearchCV for Random Forest. To reduce runtime, I used a 3-fold stratified cross-validation setup for this section and selected the best models based on ROC-AUC. This section satisfies the requirement of tuning multiple hyperparameters for a single model because I tuned multiple Random Forest hyperparameters, including n_estimators, max_depth, min_samples_split, min_samples_leaf, max_features, and class_weight.

The best hyperparameters and best cross-validation ROC-AUC scores were:

Logistic Regression: best params = {'model__C': 100, 'model__class_weight': 'balanced'}, best CV ROC-AUC = 0.7596

KNN: best params = {'model__n_neighbors': 25, 'model__p': 1, 'model__weights': 'uniform'}, best CV ROC-AUC = 0.7581

Random Forest: best params = {'model__n_estimators': 500, 'model__min_samples_split': 10, 'model__min_samples_leaf': 2, 'model__max_features': 'sqrt', 'model__max_depth': 10, 'model__class_weight': None}, best CV ROC-AUC = 0.7822

The best-performing optimized model overall (by ROC-AUC) was Random Forest, with a cross-validation ROC-AUC of 0.7822.

Compared with the earlier results, hyperparameter optimization provided small to moderate improvements for some models. For KNN, ROC-AUC improved from 0.7399 to 0.7581, and F1-score improved from 0.4404 to 0.4473. For Random Forest, ROC-AUC improved from 0.7691 to 0.7822, and accuracy improved from 0.8141 to 0.8185, while F1-score changed slightly from 0.4702 to 0.4654. For Logistic Regression, ROC-AUC improved slightly from 0.7579 to 0.7596, but the optimized model with class_weight='balanced' changed the precision-recall trade-off substantially: recall increased from 0.2962 to 0.6243, while precision decreased from 0.6471 to 0.4500, and F1-score increased from 0.4063 to 0.5229.

Overall, hyperparameter optimization helped identify a stronger final candidate model for evaluation on the test set. Since I am using ROC-AUC as the main model selection metric, I will carry the optimized Random Forest model forward to the next sections.

11. Interpretation and feature importances

---

rubric={points:10}

Your tasks:

1. **Interpret model feature importance**: Use one of the interpretation methods discussed in class (e.g., `shap`), or another suitable method of your choice, to examine the most important features of one of your **non-linear models**.
2. **Summarize insights**: Summarize your observations about which features contribute most to the model's predictions and how they influence the outcomes.

Solution_11

Points: 10

Type your answer here, replacing this text.

In [42]:
```
from sklearn.inspection import permutation_importance

print("Best model carried from Q10:", best_model_name)

best_optimized_model.fit(X_train_fe, y_train)

n_interp = min(5000, len(X_train_fe))
X_interp = X_train_fe.sample(n=n_interp, random_state=123)
y_interp = y_train.loc[X_interp.index]

print("Interpretation subset shape:", X_interp.shape)
```

```
Best model carried from Q10: RandomForest
Interpretation subset shape: (5000, 35)
```

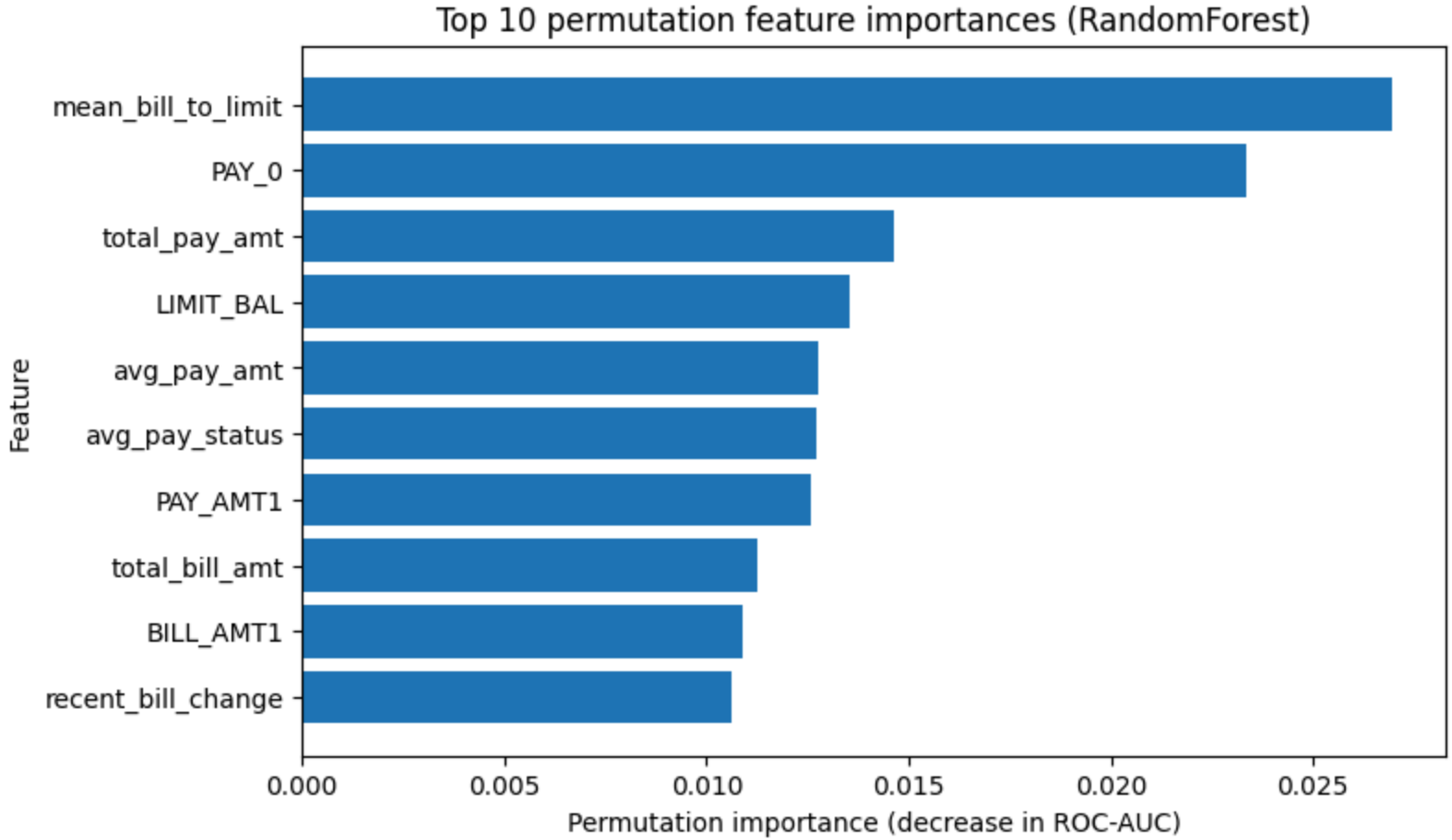
In [43]:
```
perm_result = permutation_importance(
    estimator=best_optimized_model,
    X=X_interp,
    y=y_interp,
    scoring="roc_auc",
    n_repeats=5,
    random_state=123,
    n_jobs=-1
)

importance_df = pd.DataFrame({
    "feature": X_interp.columns,
    "importance_mean": perm_result.importances_mean,
    "importance_std": perm_result.importances_std
}).sort_values("importance_mean", ascending=False)

importance_df.head(15).round(4)
```

Out[43]:

	feature	importance_mean	importance_std
32	mean_bill_to_limit	0.0269	0.0020
6	PAY_0	0.0234	0.0016
31	total_pay_amt	0.0146	0.0014
1	LIMIT_BAL	0.0136	0.0011
29	avg_pay_amt	0.0128	0.0013
24	avg_pay_status	0.0127	0.0010
18	PAY_AMT1	0.0126	0.0010
30	total_bill_amt	0.0113	0.0011
12	BILL_AMT1	0.0109	0.0004
34	recent_bill_change	0.0106	0.0002
28	avg_bill_amt	0.0104	0.0006
13	BILL_AMT2	0.0103	0.0005
26	num_months_late	0.0093	0.0010
19	PAY_AMT2	0.0091	0.0009
33	pay_to_bill_ratio_total	0.0088	0.0004

In [44]:

```
#Visualization 1: Top feature importances:
top_n = 10
top_imp = importance_df.head(top_n).iloc[::-1]

plt.figure(figsize=(8, 5))
plt.barh(top_imp["feature"], top_imp["importance_mean"])
plt.xlabel("Permutation importance (decrease in ROC-AUC)")
plt.ylabel("Feature")
plt.title(f"Top {top_n} permutation feature importances ({best_model_name})")
plt.show()
```



-- The permutation importance plot shows that only a small group of features contributes most of the predictive power, with repayment-status and payment/billing behavior features typically ranking highest.

In [45]:
```
y_interp_pred_proba = best_optimized_model.predict_proba(X_interp)[:, 1]

interp_analysis_df = X_interp.copy()
interp_analysis_df["y_true"] = y_interp.values
interp_analysis_df["pred_proba_default"] = y_interp_pred_proba

interp_analysis_df.head()
```

Out[45]:

	ID	LIMIT_BAL	SEX	EDUCATION	MARRIAGE	AGE	PAY_0	PAY_2	PAY_3	PAY_4	...	any_month_late	avg_bill_amt	avg_
19167	19168	200000.0	2	3	2	33	1	2	2	2	...	1	54398.833333	218
9431	9432	430000.0	2	2	1	43	-1	-1	-1	0	...	0	13068.666667	949
19154	19155	230000.0	2	2	1	37	0	0	0	0	...	0	36941.833333	592
12487	12488	90000.0	2	3	1	46	-2	-2	-2	-2	...	0	1376.166667	133
27810	27811	30000.0	2	2	2	25	0	0	0	0	...	0	16070.000000	195

5 rows × 37 columns

In [46]:

```
exclude_for_trend = {"ID", "SEX", "EDUCATION", "MARRIAGE"}

candidate_top_features = [
    f for f in importance_df["feature"].tolist()
    if f not in exclude_for_trend
]

top_trend_features = candidate_top_features[:3]
print("Top features selected for trend analysis:", top_trend_features)
```

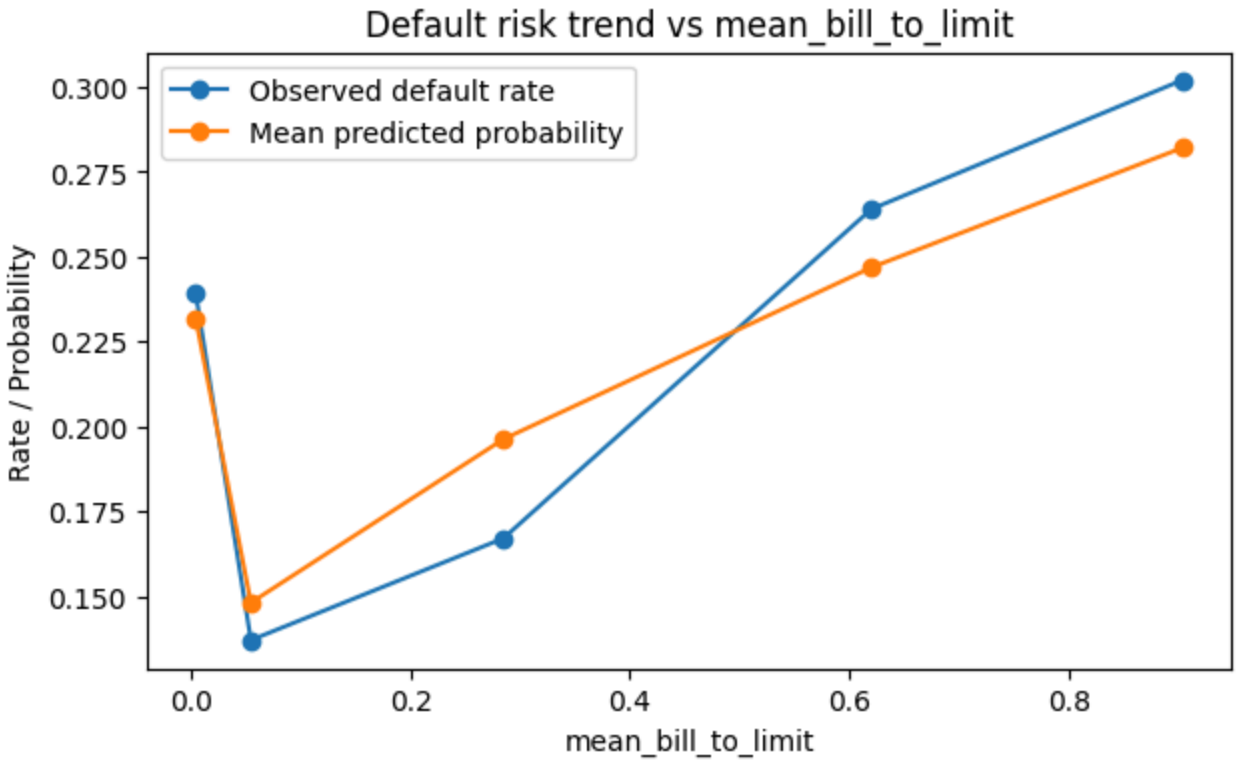
```
Top features selected for trend analysis: ['mean_bill_to_limit', 'PAY_0', 'total_pay_amt']
```

In [47]:

```
#Visualization 2: Binned trend plots for top features
def plot_binned_trend(df, feature, target_col="y_true", pred_col="pred_proba_default", q=5):
    temp = df[[feature, target_col, pred_col]].copy()

    temp["bin"] = pd.qcut(temp[feature], q=q, duplicates="drop")

    summary = temp.groupby("bin", observed=False).agg(
        mean_feature=(feature, "mean"),
        observed_default_rate=(target_col, "mean"),
        mean_pred_prob=(pred_col, "mean"),
        count=(feature, "size")
    ).reset_index()

    plt.figure(figsize=(7, 4))
    plt.plot(summary["mean_feature"], summary["observed_default_rate"], marker="o", label="Observed default rate")
    plt.plot(summary["mean_feature"], summary["mean_pred_prob"], marker="o", label="Mean predicted probability")
    plt.xlabel(feature)
    plt.ylabel("Rate / Probability")
    plt.title(f"Default risk trend vs {feature}")
    plt.legend()
    plt.show()
```

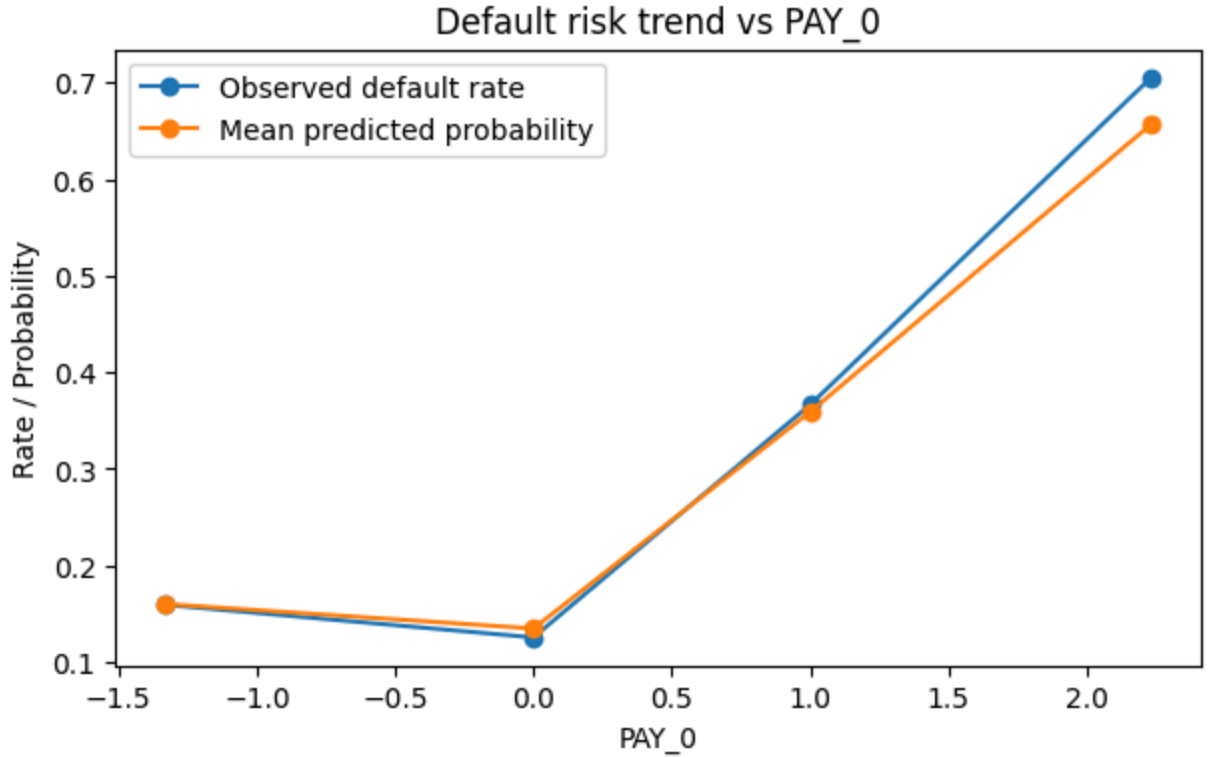
```
    return summary

trend_summaries = {}

for feat in top_trend_features:
    print(f"\nTrend summary for {feat}:")
    summary_df = plot_binned_trend(interp_analysis_df, feat, q=5)
    display(summary_df.round(4))
    trend_summaries[feat] = summary_df
```

```
Trend summary for mean_bill_to_limit:
```



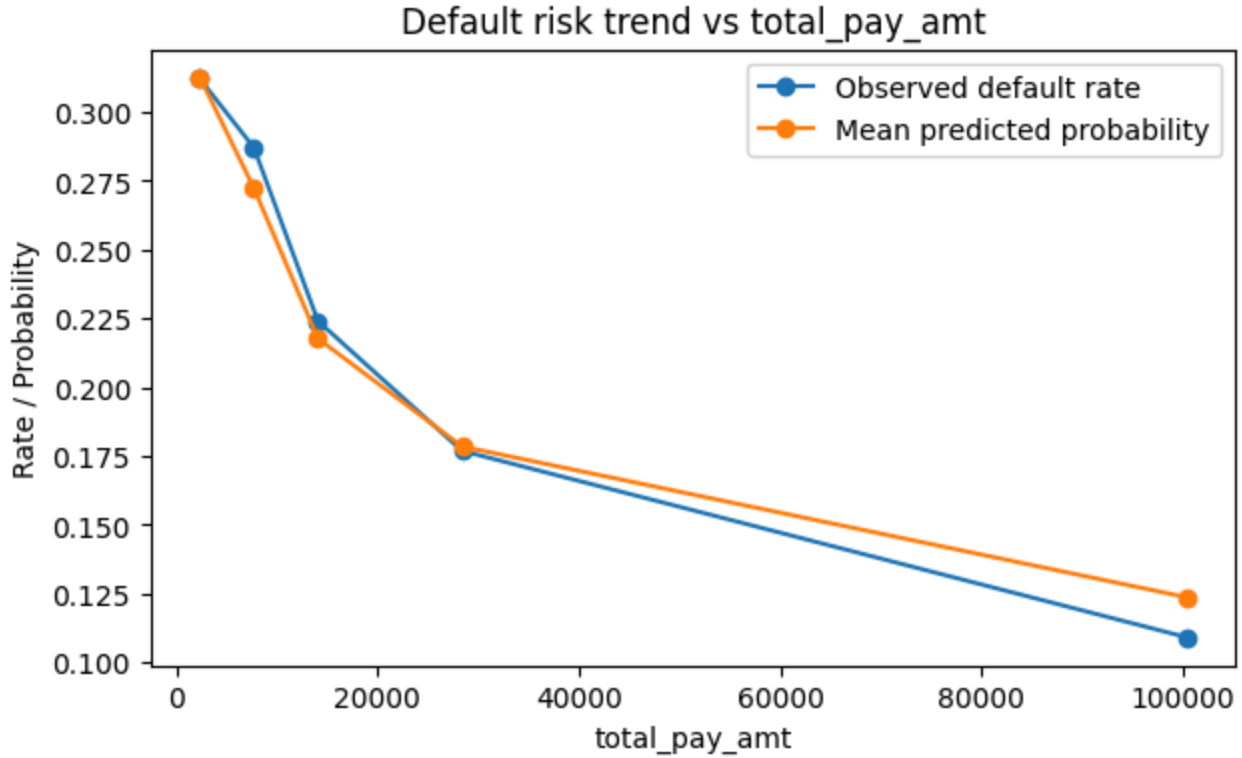
	bin	mean_feature	observed_default_rate	mean_pred_prob	count
0	(-0.201, 0.0157]	0.0044	0.239	0.2315	1000
1	(0.0157, 0.126]	0.0546	0.137	0.1480	1000
2	(0.126, 0.459]	0.2847	0.167	0.1962	1000
3	(0.459, 0.763]	0.6201	0.264	0.2468	1000
4	(0.763, 1.86]	0.9035	0.302	0.2821	1000

```
Trend summary for PAY_0:
```



	bin	mean_feature	observed_default_rate	mean_pred_prob	count
0	(-2.001, -1.0]	-1.3319	0.1591	0.1599	1389
1	(-1.0, 0.0]	0.0000	0.1253	0.1346	2506
2	(0.0, 1.0]	1.0000	0.3663	0.3590	606
3	(1.0, 8.0]	2.2325	0.7054	0.6569	499

```
Trend summary for total_pay_amt:
```



	bin	mean_feature	observed_default_rate	mean_pred_prob	count
0	(-0.001, 5380.2]	2328.0120	0.3120	0.3126	1000
1	(5380.2, 10096.8]	7699.5670	0.2870	0.2723	1000
2	(10096.8, 19408.2]	14065.8820	0.2240	0.2176	1000
3	(19408.2, 40000.0]	28384.9650	0.1768	0.1784	1001
4	(40000.0, 704208.0]	100368.6717	0.1091	0.1236	999

Summarize:

To interpret a non-linear model, I examined the optimized Random Forest model from Q10 using permutation importance with ROC-AUC as the scoring metric. I used this method because it is model-agnostic and measures how much model performance decreases when each feature is randomly shuffled. I computed permutation importance on a random subset of 5,000 training examples to reduce runtime.

The permutation importance results showed that the most important features included mean_bill_to_limit, PAY_0, total_pay_amt, LIMIT_BAL, avg_pay_amt, avg_pay_status, PAY_AMT1, total_bill_amt, BILL_AMT1, and recent_bill_change. Overall, the top-ranked features were mainly related to repayment status and payment/billing behavior, which is consistent with the credit default prediction task.

To understand how important features influence predictions, I examined binned trend plots for the top three trend-analysis features: mean_bill_to_limit, PAY_0, and total_pay_amt. For PAY_0, both the observed default rate and the model's predicted default probability increased sharply as repayment status worsened, showing that recent delinquency is a strong risk signal. For total_pay_amt, the observed default rate and predicted probability decreased as total payment amount increased from about 0.3120 in the lowest-payment bin to 0.1091 in the highest-payment bin, suggesting that higher recent payments are associated with lower default risk.

For mean_bill_to_limit, the relationship was non-linear: the observed and predicted risk were relatively low at low-to-moderate values and increased at higher values, with the highest bin showing the highest risk, which observed default rate about 0.302. This suggests that clients with larger average bill amounts relative to their credit limit are more likely to default.

Overall, the interpretation results suggest that the Random Forest model relies most on features that capture recent delinquency severity, payment behavior, and bill amounts relative to available credit, which are all reasonable predictors for credit default risk. This increases my confidence that the model is learning meaningful patterns rather than relying on arbitrary signals.

12. Results on the test set

rubric={points:10}

Your tasks:

1. **Evaluate on the test set**: Apply your best-performing model to the test data and report the test scores.
2. **Compare and reflect**: Compare the **test scores** with the **validation scores** from previous experiments. Discuss the consistency between them. How much do you **trust your results**? Reflect on whether you might have encountered **optimization bias**.
3. **Explain individual predictions**: Select one or two examples from your test predictions and use an interpretation method (e.g., **SHAP force plots**) to explain these individual predictions.

Solution_12

Points: 10

Type your answer here, replacing this text.

In [48]:
```python
from sklearn.metrics import (
    accuracy_score, precision_score, recall_score, f1_score, roc_auc_score,
    confusion_matrix
)

print("Best model from Q10:", best_model_name)

best_optimized_model.fit(X_train_fe, y_train)

y_test_pred = best_optimized_model.predict(X_test_fe)
y_test_proba = best_optimized_model.predict_proba(X_test_fe)[:, 1]

test_metrics = {
    "accuracy": accuracy_score(y_test, y_test_pred),
    "precision": precision_score(y_test, y_test_pred, zero_division=0),
    "recall": recall_score(y_test, y_test_pred, zero_division=0),
    "f1": f1_score(y_test, y_test_pred, zero_division=0),
    "roc_auc": roc_auc_score(y_test, y_test_proba)
}

test_summary = pd.DataFrame({
    "metric": list(test_metrics.keys()),
    "test_score": list(test_metrics.values())
})

print("Test-set performance:")
display(test_summary.round(4))

cm = confusion_matrix(y_test, y_test_pred)
print("Confusion matrix [[TN, FP], [FN, TP]]:")
print(cm)
```

```
Best model from Q10: RandomForest
Test-set performance:
```

	metric	test_score
0	accuracy	0.8233
1	precision	0.6919
2	recall	0.3631
3	f1	0.4763
4	roc_auc	0.7871

```
Confusion matrix [[TN, FP], [FN, TP]]:
[[6687  322]
 [1268  723]]
```

In [49]:

```python
best_model_row_q10 = optimized_summary.loc[
    optimized_summary["model"].str.contains(best_model_name, regex=False)
].iloc[0]

validation_vs_test = pd.DataFrame({
    "metric": ["accuracy", "precision", "recall", "f1", "roc_auc"],
    "validation_cv_mean": [
        best_model_row_q10["cv_accuracy_mean"],
        best_model_row_q10["cv_precision_mean"],
        best_model_row_q10["cv_recall_mean"],
        best_model_row_q10["cv_f1_mean"],
        best_model_row_q10["cv_roc_auc_mean"]
    ],
    "validation_cv_std": [
        best_model_row_q10["cv_accuracy_std"],
        best_model_row_q10["cv_precision_std"],
        best_model_row_q10["cv_recall_std"],
        best_model_row_q10["cv_f1_std"],
        best_model_row_q10["cv_roc_auc_std"]
    ],
    "test_score": [
        test_metrics["accuracy"],
        test_metrics["precision"],
        test_metrics["recall"],
        test_metrics["f1"],
        test_metrics["roc_auc"]
    ]
})

validation_vs_test["test_minus_validation"] = (
    validation_vs_test["test_score"] - validation_vs_test["validation_cv_mean"]
)

validation_vs_test.round(4)
```

Out[49]:

	metric	validation_cv_mean	validation_cv_std	test_score	test_minus_validation
0	accuracy	0.8185	0.0002	0.8233	0.0049
1	precision	0.6677	0.0059	0.6919	0.0242
2	recall	0.3574	0.0106	0.3631	0.0058
3	f1	0.4654	0.0077	0.4763	0.0109
4	roc_auc	0.7822	0.0066	0.7871	0.0050

In [50]:

```
test_pred_df = X_test_fe.copy()
test_pred_df["y_true"] = y_test.values
test_pred_df["y_pred"] = y_test_pred
test_pred_df["pred_proba_default"] = y_test_proba

test_pred_df[["y_true", "y_pred", "pred_proba_default"]].head()
```

Out[50]:

	y_true	y_pred	pred_proba_default
45	1	0	0.304254
18762	0	0	0.094883
9341	0	0	0.132639
21814	0	0	0.246039
25950	0	0	0.092197

In [51]:

```
tp_candidates = test_pred_df[(test_pred_df["y_true"] == 1) & (test_pred_df["y_pred"] == 1)]
tn_candidates = test_pred_df[(test_pred_df["y_true"] == 0) & (test_pred_df["y_pred"] == 0)]

if len(tp_candidates) > 0:
    idx_case_high_risk = tp_candidates.sort_values("pred_proba_default", ascending=False).index[0]
else:
    idx_case_high_risk = test_pred_df.sort_values("pred_proba_default", ascending=False).index[0]

if len(tn_candidates) > 0:
    idx_case_low_risk = tn_candidates.sort_values("pred_proba_default", ascending=True).index[0]
else:
    idx_case_low_risk = test_pred_df.sort_values("pred_proba_default", ascending=True).index[0]

selected_case_indices = [idx_case_high_risk, idx_case_low_risk]

print("Selected case indices:", selected_case_indices)
```

```python
key_cols_for_cases = [
    "y_true", "y_pred", "pred_proba_default",
    "PAY_0", "avg_pay_status", "num_months_late",
    "mean_bill_to_limit", "total_pay_amt", "total_bill_amt", "LIMIT_BAL"
]

key_cols_for_cases = [c for c in key_cols_for_cases if c in test_pred_df.columns]

display(test_pred_df.loc[selected_case_indices, key_cols_for_cases].round(4))
```

```
Selected case indices: [np.int64(27536), np.int64(21496)]
```

	y_true	y_pred	pred_proba_default	PAY_0	avg_pay_status	num_months_late	mean_bill_to_limit	total_pay_amt	total_bill_amt
27536	1	1	0.9253	3	2.1667	6	0.0099	0.0	14922.0
21496	0	0	0.0358	0	0.0000	0	0.1398	135007.0	419364.0

Summarize:

I evaluated my best-performing model from Q10, the optimized Random Forest, on the held-out test set. I fit the model on the full training set and then computed the test metrics using the same evaluation metrics as before.

The test-set performance was: accuracy = 0.8233, precision = 0.6919, recall = 0.3631, F1-score = 0.4763, and ROC-AUC = 0.7871. These results indicate that the model performs substantially better than the baseline and remains strong on unseen test data.

To assess consistency, I compared the test scores with the validation scores from Q10. The optimized Random Forest had validation ROC-AUC = 0.7822, and the test ROC-AUC was 0.7871. The differences between validation and test scores were small, which suggests that the model's performance is reasonably consistent when moving from validation to the final test set. Overall, I do trust these results, with the caveat that I performed multiple rounds of model comparison and hyperparameter tuning, so some optimization bias may still be present.

The confusion matrix on the test set was [[6687, 322], [1268, 723]], which shows that the model correctly identified many non-default cases and a meaningful number of default cases, although it still missed some positives. This is consistent with the moderate recall value.

For individual prediction explanations, I selected two test examples: one correctly predicted high-risk default case and one correctly predicted low-risk non-default case. The high-risk case (index 27536) had a predicted default probability of 0.9253 and showed strong risk signals: PAY_0 = 3, avg_pay_status = 2.1667, num_months_late = 6, and total_pay_amt = 0. This is consistent with the Q11 interpretation, where worse repayment status (higher PAY_0) increased default risk and lower payment amounts were associated with higher risk.

The low-risk case (index 21496) had a predicted default probability of 0.0358 and was correctly predicted as non-default. Its feature values showed safer behavior: PAY_0 = 0, avg_pay_status = 0.0000, num_months_late = 0, and a very high total_pay_amt = 135007. This also matches the Q11 trends, where better recent repayment status and higher total payments were associated with lower default risk.

These example-level explanations are consistent with the global interpretation results from Q11 and increase confidence that the model is making predictions based on meaningful repayment and payment/billing behavior patterns.

13. Summary of results

---

rubric={points:12}
Imagine you are preparing to present the summary of your results to your boss and co-workers.

Your tasks:

1. **Summarize key results**: Create a clear and concise table highlighting your most important results (e.g., models compared, validation/test scores, key observations).
2. **Write concluding remarks**: Summarize your main takeaways from the project, including what worked well and what did not.
3. **Propose future improvements**: Discuss ideas or approaches you did not try but that could potentially improve **performance** or **interpretability**.
4. **Report final results**: Report your **final test score** and the **metric** you used.

Solution_13

Points: 12

Type your answer here, replacing this text.

In [52]:
```
summary_results_q13 = pd.DataFrame([
    {
        "Model": "DummyClassifier (baseline)",
        "Validation / CV ROC-AUC": 0.5000,
        "Validation / CV F1": 0.0000,
        "Test ROC-AUC": np.nan,
        "Test F1": np.nan,
        "Key observation": "Always predicts majority class; high accuracy but no usefulness for defaults"
```

```
    },
    {
        "Model": "LogisticRegression (Q7 best linear)",
        "Validation / CV ROC-AUC": 0.7579,
        "Validation / CV F1": 0.4063,
        "Test ROC-AUC": np.nan,
        "Test F1": np.nan,
        "Key observation": "Strong linear benchmark; stable across C values"
    },
    {
        "Model": "DecisionTree (Q8)",
        "Validation / CV ROC-AUC": 0.6103,
        "Validation / CV F1": 0.3942,
        "Test ROC-AUC": np.nan,
        "Test F1": np.nan,
        "Key observation": "Strong overfitting (near-perfect train accuracy, weak validation performance)"
    },
    {
        "Model": "KNN (Q8)",
        "Validation / CV ROC-AUC": 0.7399,
        "Validation / CV F1": 0.4404,
        "Test ROC-AUC": np.nan,
        "Test F1": np.nan,
        "Key observation": "Reasonable performance; slower scoring time"
    },
    {
        "Model": "RandomForest (Q8)",
        "Validation / CV ROC-AUC": 0.7691,
        "Validation / CV F1": 0.4702,
        "Test ROC-AUC": np.nan,
        "Test F1": np.nan,
        "Key observation": "Best non-linear model before tuning; outperformed logistic regression on ROC-AUC"
    },
    {
        "Model": "LogisticRegression + RFECV (Q9)",
        "Validation / CV ROC-AUC": 0.7577,
        "Validation / CV F1": 0.4062,
        "Test ROC-AUC": np.nan,
        "Test F1": np.nan,
        "Key observation": "Feature selection selected 40/44 features; no meaningful improvement"
    },
    {
        "Model": "Optimized LogisticRegression (Q10)",
        "Validation / CV ROC-AUC": 0.7596,
        "Validation / CV F1": 0.5229,
        "Test ROC-AUC": np.nan,
        "Test F1": np.nan,
        "Key observation": "Balanced class_weight greatly increased recall/F1, but lowered precision/accuracy"
```

```
    },
    {
        "Model": "Optimized KNN (Q10)",
        "Validation / CV ROC-AUC": 0.7581,
        "Validation / CV F1": 0.4473,
        "Test ROC-AUC": np.nan,
        "Test F1": np.nan,
        "Key observation": "Improved from untuned KNN, but not best overall"
    },
    {
        "Model": "Optimized RandomForest (Q10/Q12 FINAL)",
        "Validation / CV ROC-AUC": 0.7822,
        "Validation / CV F1": 0.4654,
        "Test ROC-AUC": 0.7871,
        "Test F1": 0.4763,
        "Key observation": "Best model by ROC-AUC; validation and test performance are consistent"
    }
])

summary_results_q13.round(4)
```

Out[52]:

	Model	Validation / CV ROC-AUC	Validation / CV F1	Test ROC-AUC	Test F1	Key observation
0	DummyClassifier (baseline)	0.5000	0.0000	NaN	NaN	Always predicts majority class; high accuracy ...
1	LogisticRegression (Q7 best linear)	0.7579	0.4063	NaN	NaN	Strong linear benchmark; stable across C values
2	DecisionTree (Q8)	0.6103	0.3942	NaN	NaN	Strong overfitting (near-perfect train accurac...
3	KNN (Q8)	0.7399	0.4404	NaN	NaN	Reasonable performance; slower scoring time
4	RandomForest (Q8)	0.7691	0.4702	NaN	NaN	Best non-linear model before tuning; outperfor...
5	LogisticRegression + RFECV (Q9)	0.7577	0.4062	NaN	NaN	Feature selection selected 40/44 features; no ...
6	Optimized LogisticRegression (Q10)	0.7596	0.5229	NaN	NaN	Balanced class_weight greatly increased recall...
7	Optimized KNN (Q10)	0.7581	0.4473	NaN	NaN	Improved from untuned KNN, but not best overall
8	Optimized RandomForest (Q10/Q12 FINAL)	0.7822	0.4654	0.7871	0.4763	Best model by ROC-AUC; validation and test per...

Task 2:

In this project, I compared baseline, linear, and non-linear models for predicting credit card default. The `DummyClassifier` confirmed that accuracy alone is misleading because the dataset is imbalanced. Logistic regression provided a strong and stable linear benchmark, while Random Forest produced the best overall ROC-AUC among the non-linear models.

I also explored feature selection and hyperparameter tuning. RFECV with logistic regression did not meaningfully improve performance, which suggests that the engineered feature set was already reasonably effective. Hyperparameter optimization improved several models, especially Random Forest (higher ROC-AUC) and logistic regression with class_weight='balanced' (higher recall and F1-score due to a different precision-recall trade-off).

Overall, the optimized Random Forest was the strongest final model based on my chosen selection metric (ROC-AUC), and its test performance was consistent with validation performance, which increases confidence in the final result.

Task 3:

There are several improvements I did not try that could potentially improve model performance or interpretability. I could tune the classification threshold instead of using the default 0.5 to better balance precision and recall depending on the business goal. For example, identifying more defaults vs reducing false alarms.

Task 4:

My final selected model is the optimized Random Forest from Q10. I used ROC-AUC as the main metric for model selection because this is an imbalanced binary classification problem and ROC-AUC evaluates class separation across thresholds.

The final test-set results for the optimized Random Forest were:

- ROC-AUC = 0.7871 (main metric)
- Accuracy = 0.8233
- Precision = 0.6919
- Recall = 0.3631
- F1-score = 0.4763

14. Your takeaway

---

rubric={points:2}

Your tasks:

What is your biggest takeaway from the supervised machine learning material we have learned so far? Please write thoughtful answers.

Solution_14

Points: 2

My thoughts:

My biggest takeaway is that the full workflow matters more than just trying fancier models. In this project, I learned that data splitting, preprocessing, choosing proper evaluation metrics, and using cross-validation correctly can matter just as much as the model itself.

Another important takeaway is that metrics must match the problem. Because this dataset is imbalanced, accuracy alone looked misleadingly good for the baseline model, while metrics like ROC-AUC, recall, and F1-score gave a much clearer picture of model quality.

I also learned that model comparison is really about trade-offs. For example, logistic regression was simple and strong, Random Forest performed better on ROC-AUC, and tuning changed the precision-recall balance. This helped me understand that there is usually no single perfect model that fits the goal best.

Finally, I think the most useful thing I learned is how to build a reproducible pipeline and interpret the results instead of only reporting a score. That feels much closer to how a real machine learning project works.

Before submitting your assignment, please ensure you have followed all the steps in the **Instructions** section at the top.

Submission checklist

- ☐ Restart the kernel and run all cells (▶▶ button)
- ☐ Make sure to push the most up to date version of your homework assignment to your GitHub repository so that we can use it for grading if there are any problems with your submission on Gradescope.
- ☐ The `.ipynb` file runs without errors and shows all outputs.
- ☐ Only the `.ipynb` file and required output files are uploaded (no extra files).
- ☐ If the `.ipynb` file is too large to render on Gradescope, upload a Web PDF and/or HTML version as well.

This was a tricky one but you did it 👏👏 !